



Ayushasthra

Ayurvedic Medicine Identification
and Health Management App

**Group Number :
R25-068**



Introduction

Our Ayurvedic Medicine Identification and Health Management App combines the ancient wisdom of Ayurveda with modern technology.

By leveraging

- > Artificial intelligence.**
- > Image processing**
- > Data analytics**

This app makes

- > Ayurvedic care more accessible and personalized.**
- > Helps users identify medicinal plants**
- > Gain health insights**

By focusing demand for holistic and natural health solutions, this app offers an accurate, user-friendly way to bridge traditional practices and modern needs.

Mainly Focusing on,

- > Tongue Disease identification and treatment suggestions**
- > Skin disease identification and treatment Suggestions.**
- > Ayurvedic medicinal plant identification with detailed usage guidelines.**
- > Personalized yoga and lifestyle guidance based on wearable health data.**

Research Questions

- Ayurvedic diagnosis is limited by practitioner dependency, delaying treatment and reducing accessibility.

Skin diseases lack integrated solutions combining Ayurvedic and modern approaches. Health tools fail to offer personalized care by excluding Ayurvedic data, while difficulties in identifying medicinal plants further restrict remedy access.

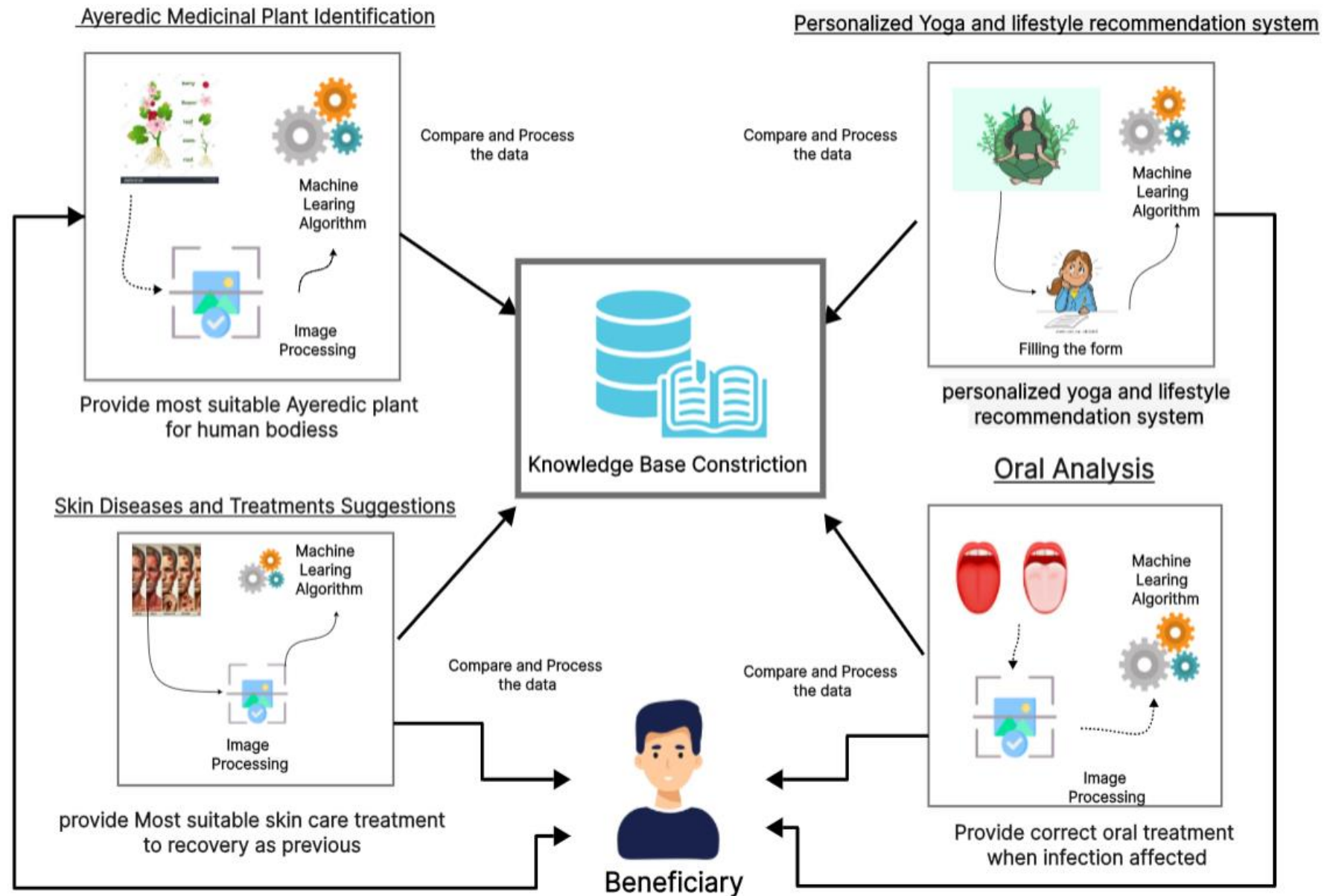


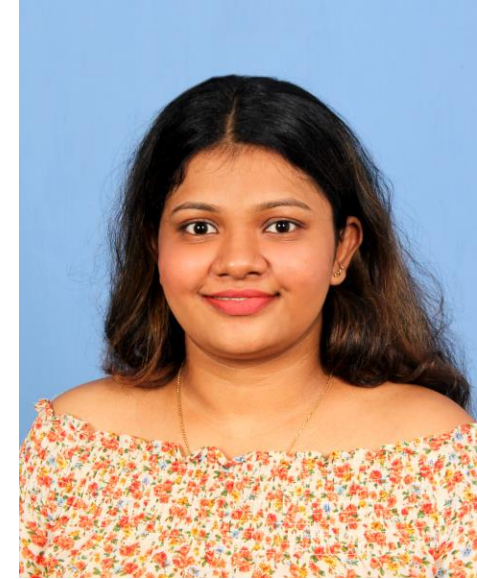
Objectives

- The **Primary Goal** is to create an AI-driven mobile app that combines Ayurvedic principles with modern technology to provide personalized health management and treatment recommendations.
- The objectives include
 - > Tongue imaging analysis for personalized Ayurvedic advice
 - > Skin disease identification framework blending Ayurvedic and modern diagnostics
 - > Tailored yoga and lifestyle recommendations using wearable health data
 - > Medicinal plant identification with detailed usage guidance.



System Diagram





PEIRIS P.H.R - IT21584718

AI-POWERED TONGUE DIAGNOSTICS FOR HEALTH MONITORING, WELLNESS GUIDANCE AND NUTRITION SUGGESTING

Information Technology (IT)

Introduction

Background



- **Tongue diagnosis in Ayurveda:** The condition of the tongue (its color, texture, and coating) is used in Ayurvedic healthcare to assess overall health.
 1. **Detects health issues:** Identify problems like dehydration, nutrient deficiencies, and inflammation.
 2. **AI enhances accuracy :** detecting subtle changes in tongue images.
 3. **Combining AI with Ayurveda Home Remedies:** Integrating AI with Ayurvedic methods makes health assessments more personalized and accessible.
 4. **Reduces clinic visits:** AI-powered tools allow users to monitor their health at home,

Features Comparison Table

Features	Research 1 [1]	Research 2 [2]	Research 3 [3]	Research 4 [4]	Proposed Solution
User Friendly	Low	Low	Medium	Low	High
Mobile Solution	✗	✗	✗	✗	✓
Integration of AI with Ayurvedic Tongue Diagnostics	✗	✗	✗	✓	✓
Innovative Image Processing for Tongue Analysis (Color, Texture, Hydration)	✗	✗	✓	✓	✓
Recovery Tracking System	✗	✗	✗	✗	✓
Recovery tracking system and progress monitoring					
Real-Time Health Risk Assessment	✗	✗	✗	✗	✓
Varied AI Model: CNNs, Computer Vision, ML for Disease Detection	✗	✗	✗	✓	✓

Selected Diseases

Black hairy diseases	Fissured Tongue



Black hairy tongue is caused by a buildup of dead skin cells on the tiny bumps on the tongue called papillae



Fissured tongue is a harmless condition that causes grooves or furrows on the tongue



Research Gap

Limited AI Integration in Tongue Diagnosis – Existing healthcare apps do not fully utilize AI for accurate tongue-based health assessments.

No Personalized Wellness Tracking – Most apps lack personalized progress tracking based on tongue health changes.

Lack of Digital Ayurvedic Insights – Ayurvedic tongue diagnosis is not well-represented in modern health apps.

Insufficient Tongue Image Datasets – There is a shortage of annotated tongue images linked to different health conditions.

Limited Access to Remote Health Monitoring – People, especially in rural areas, struggle to access Ayurvedic experts, needing AI-driven remote solutions.

Requirements Specification

Functional Requirements

- **Health Risk Detection and Staging**

- The app must analyze tongue images to **detect health conditions** (e.g., dehydration, inflammation, infections) and classify them into **mild, moderate, or severe stages** using AI.

- **Personalized Wellness Recommendations**

- Instead of direct treatment, the app should provide **general Ayurvedic-based wellness advice**, including **hydration tips, dietary suggestions, and lifestyle recommendations** based on the detected health condition.

Non - Functional Requirements

- **Data Security and Privacy.**

Ensure secure storage and processing of user data, complying with privacy regulations and safeguarding sensitive health information.

- **Performance.**

The app should deliver real-time results, with image analysis and treatment recommendations completed within a few seconds.

Research Problems

- **No scalable AI-based tongue diagnosis tools** exist for health assessments.
- **Challenges in combining AI with Ayurvedic principles** for accurate analysis.
- **Lack of sufficient and structured tongue image datasets** for AI training.
- **No personalized health tracking** based on tongue condition changes.
- **Limited access to AI-driven Ayurvedic health solutions**, especially in rural areas.



Research Objectives

Specific Objective

- Develop an AI- Powered mobile application to detect diseases on human tongue and suggest home ramadies and nutritions that humen need until they recovery.

Sub Objectives

- Collect a diverse dataset of tongue images** labeled with different health conditions (e.g., normal, dehydrated, inflamed, and infected) to improve AI accuracy.
- Train and deploy a deep learning model** to analyze tongue images, detect health risks, and track changes over time.
- Develop a personalized wellness recommendation system** based on AI analysis and Ayurvedic insights to suggest hydration, nutrition, and lifestyle improvements.
- Integrate a notification system** to remind users about hydration, nutrition intake, and health progress monitoring.
- Ensure secure data storage and privacy measures** to protect user health information.

Achievement



- Successful data collection for model training
- Trained two models for Selected Diseases
- Highest accuracy models up to now
- Developed the mobile application (Emulator)

Fissured tongue

- A **fissured tongue** is when the tongue has deep grooves or cracks on its surface.
- It's usually harmless and can happen as
 - people get older.
 - food or bacteria can get stuck in these cracks, which might cause irritation or bad breath. Cleaning the tongue regularly can help keep it clean and reduce any issues.





Stages of Fissured tongue

1. Successful data collection for model training

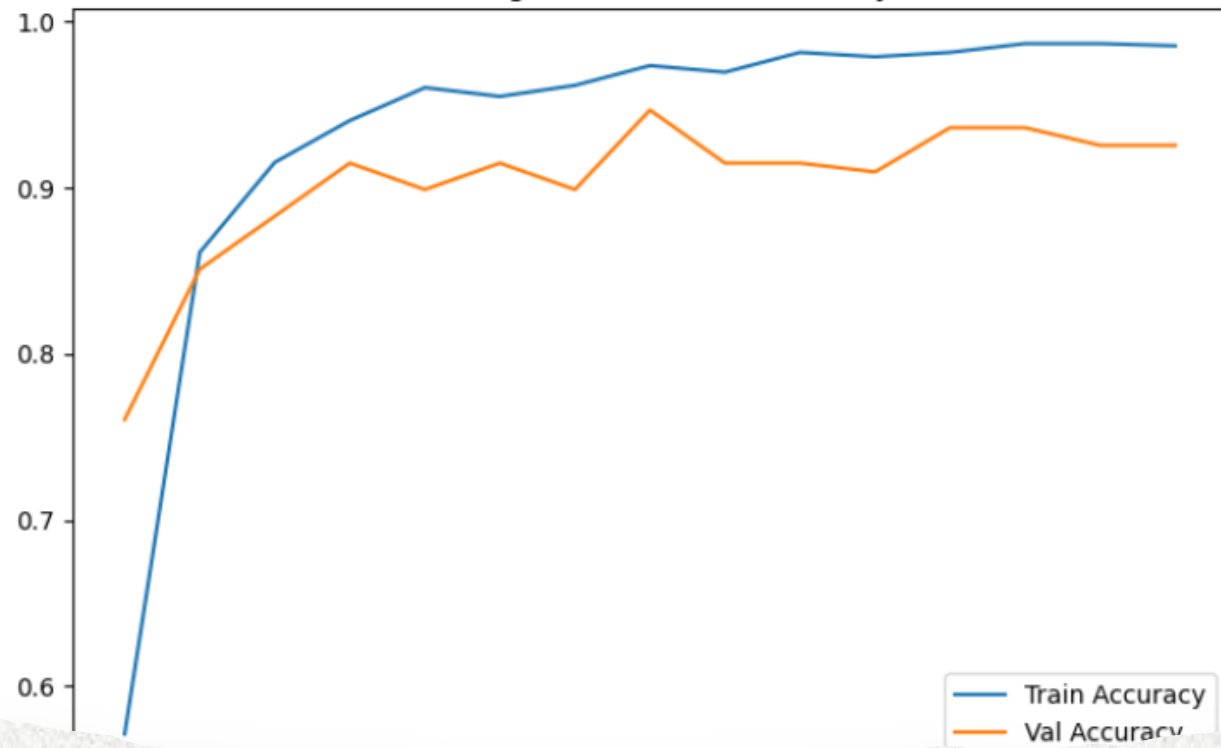


Accuracy

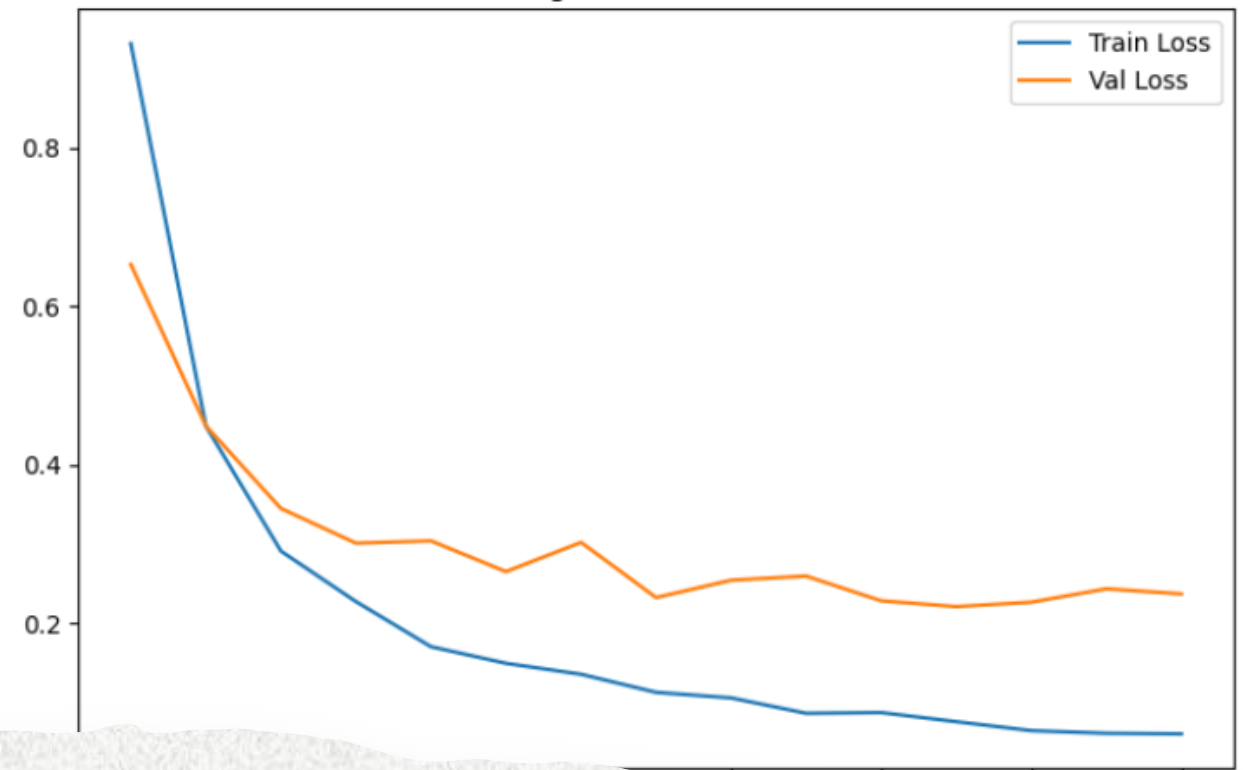
- Accuracy
97%

```
self._warn_if_super_not_called()
Epoch 1/15
24/24 ————— 287s 12s/step - accuracy: 0.4273 - loss: 1.1334 - val_accuracy: 0.7606 - val_loss: 0.6530
Epoch 2/15
24/24 ————— 16s 650ms/step - accuracy: 0.8231 - loss: 0.5062 - val_accuracy: 0.8511 - val_loss: 0.4492
Epoch 3/15
24/24 ————— 20s 641ms/step - accuracy: 0.9114 - loss: 0.3028 - val_accuracy: 0.8830 - val_loss: 0.3452
Epoch 4/15
24/24 ————— 15s 632ms/step - accuracy: 0.9284 - loss: 0.2507 - val_accuracy: 0.9149 - val_loss: 0.3015
Epoch 5/15
24/24 ————— 15s 641ms/step - accuracy: 0.9613 - loss: 0.1757 - val_accuracy: 0.8989 - val_loss: 0.3043
Epoch 6/15
24/24 ————— 17s 735ms/step - accuracy: 0.9450 - loss: 0.1526 - val_accuracy: 0.9149 - val_loss: 0.2655
Epoch 7/15
24/24 ————— 15s 631ms/step - accuracy: 0.9570 - loss: 0.1384 - val_accuracy: 0.8989 - val_loss: 0.3024
Epoch 8/15
24/24 ————— 17s 740ms/step - accuracy: 0.9668 - loss: 0.1224 - val_accuracy: 0.9468 - val_loss: 0.2328
Epoch 9/15
24/24 ————— 18s 741ms/step - accuracy: 0.9691 - loss: 0.1062 - val_accuracy: 0.9149 - val_loss: 0.2547
Epoch 10/15
24/24 ————— 16s 663ms/step - accuracy: 0.9731 - loss: 0.0941 - val_accuracy: 0.9149 - val_loss: 0.2599
Epoch 11/15
24/24 ————— 15s 639ms/step - accuracy: 0.9776 - loss: 0.0852 - val_accuracy: 0.9096 - val_loss: 0.2288
Epoch 12/15
24/24 ————— 15s 636ms/step - accuracy: 0.9789 - loss: 0.0772 - val_accuracy: 0.9362 - val_loss: 0.2214
Epoch 13/15
...
Epoch 14/15
24/24 ————— 15s 644ms/step - accuracy: 0.9854 - loss: 0.0641 - val_accuracy: 0.9255 - val_loss: 0.2438
Epoch 15/15
24/24 ————— 15s 631ms/step - accuracy: 0.9791 - loss: 0.0692 - val_accuracy: 0.9255 - val_loss: 0.2375
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Training and Validation Accuracy

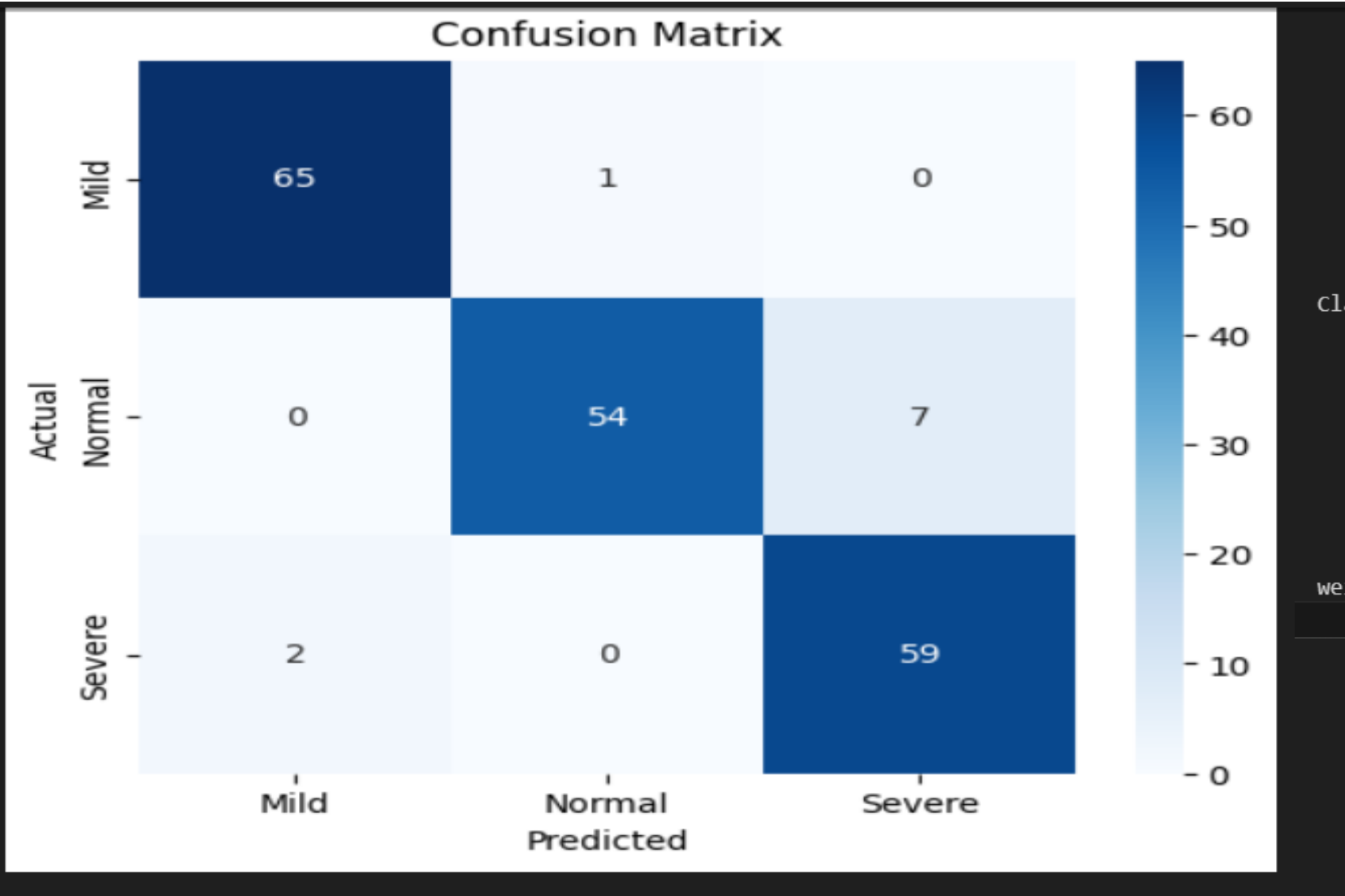


Training and Validation Loss



Testing Validation Accuracy & Lost

Confusion Matrix



Classification Report:

	precision	recall	f1-score	support
Mild	0.97	0.98	0.98	66
Normal	0.98	0.89	0.93	61
Severe	0.89	0.97	0.93	61
accuracy			0.95	188
macro avg	0.95	0.95	0.95	188
weighted avg	0.95	0.95	0.95	188

Layers

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 224, 224, 3)	0	-
Conv1 (Conv2D)	(None, 112, 112, 32)	864	input_layer[0][0]
bn_Conv1 (BatchNormalization)	(None, 112, 112, 32)	128	Conv1[0][0]
Conv1_relu (ReLU)	(None, 112, 112, 32)	0	bn_Conv1[0][0]
expanded_conv_depthwise (DepthwiseConv2D)	(None, 112, 112, 32)	288	Conv1_relu[0][0]
expanded_conv_depthwise_... (BatchNormalization)	(None, 112, 112, 32)	128	expanded_conv_depthwi...
expanded_conv_depthwise_... (ReLU)	(None, 112, 112, 32)	0	expanded_conv_depthwi...
expanded_conv_project (Conv2D)	(None, 112, 112, 16)	512	expanded_conv_depthwi...
expanded_conv_project_BN (BatchNormalization)	(None, 112, 112, 16)	64	expanded_conv_project...
block_1_expand (Conv2D)	(None, 112, 112, 96)	1,536	expanded_conv_project...
block_1_expand_BN (BatchNormalization)	(None, 112, 112, 96)	384	block_1_expand[0][0]
block_1_expand_relu (ReLU)	(None, 112, 112, 96)	0	block_1_expand_BN[0][...]
block_1_pad (ZeroPadding2D)	(None, 113, 113, 96)	0	block_1_expand_relu[0...
block_1_depthwise (DepthwiseConv2D)	(None, 56, 56, 96)	864	block_1_pad[0][0]

Black Hairy Tongue

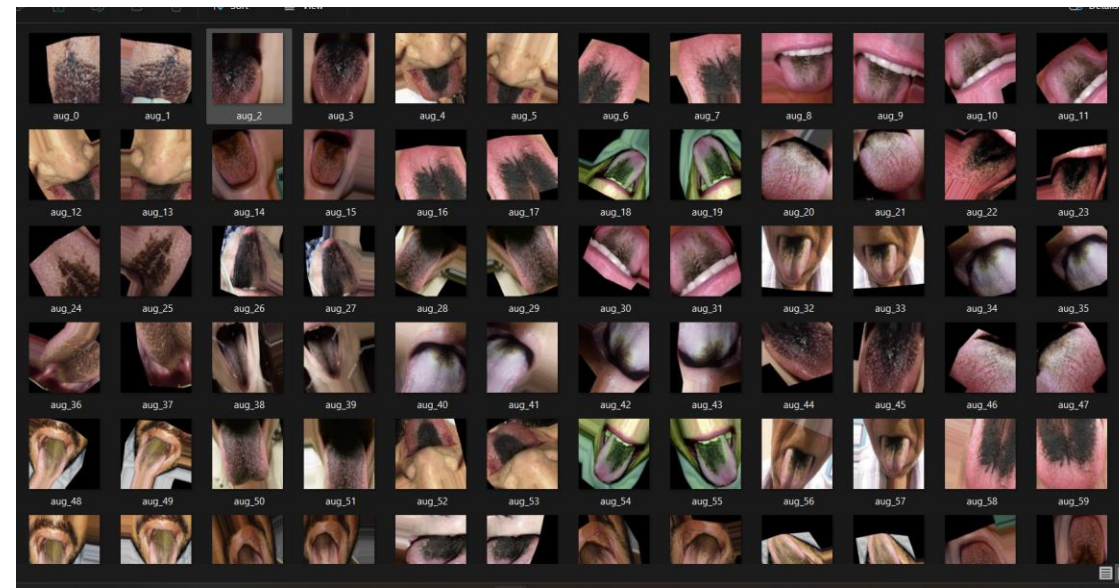
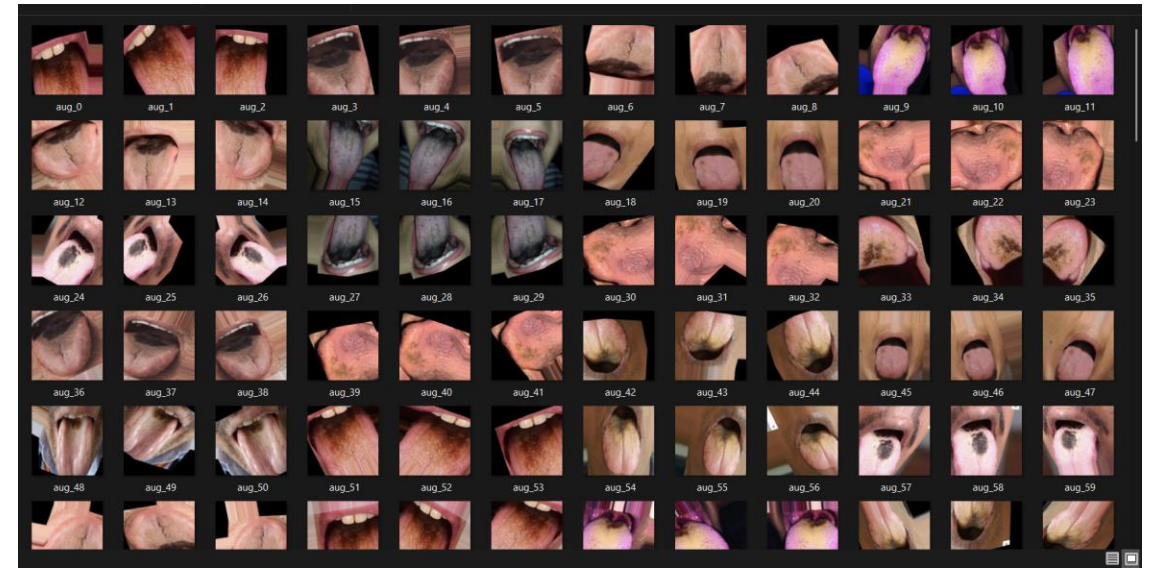
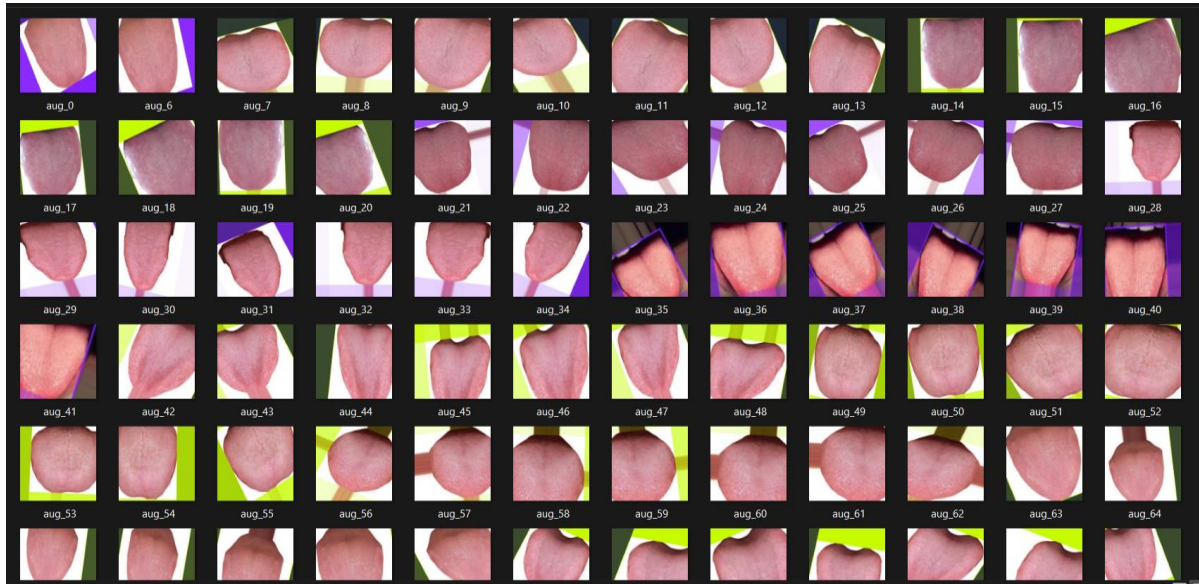
- **Black hairy tongue** is a condition where the tongue looks dark and fuzzy due to buildup of dead skin cells and bacteria. It's usually harmless and can be caused by poor oral hygiene, smoking, or certain medications. Regular brushing can help clear it up.





Stages of Black hairy

1.Successful data collection for model training

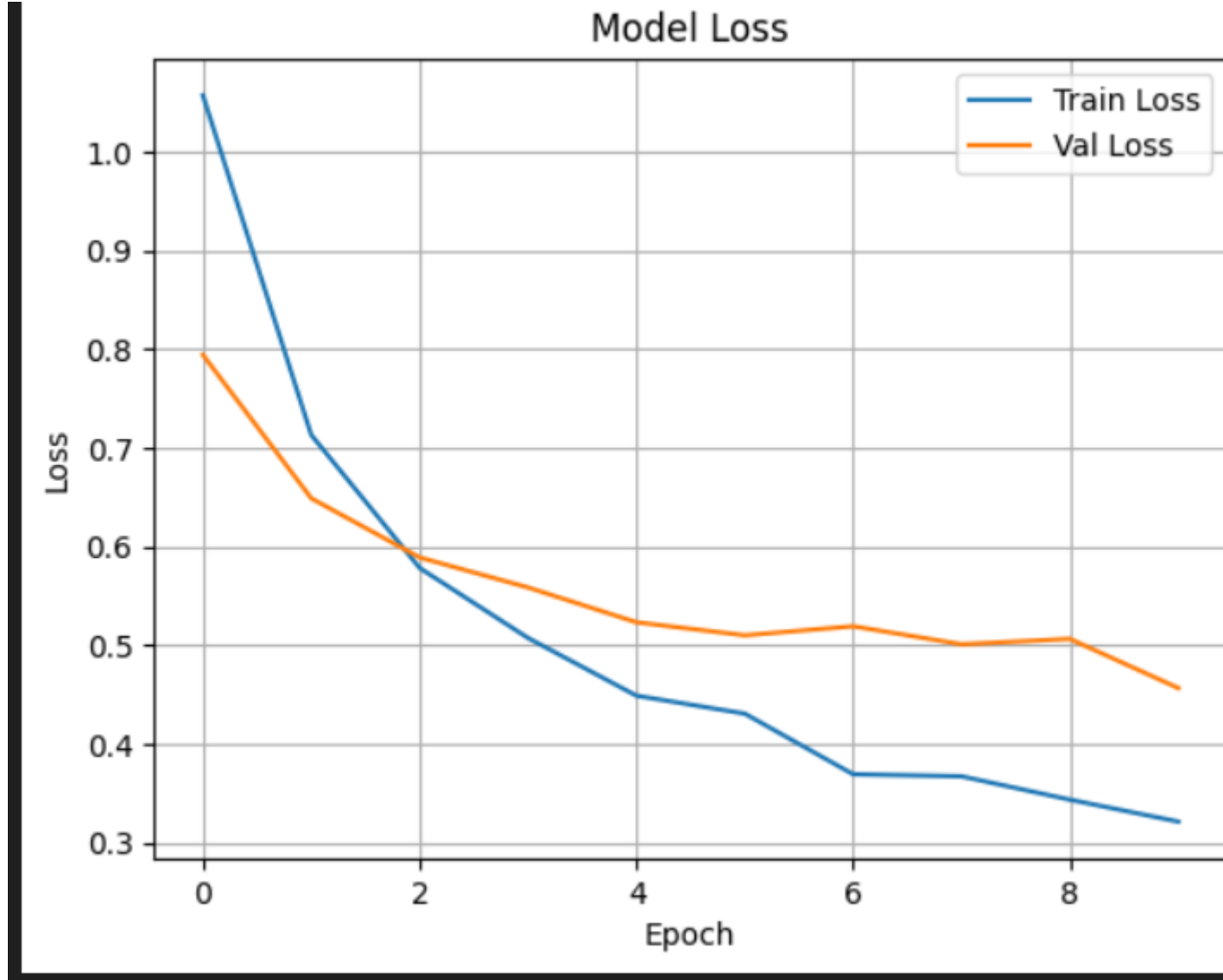
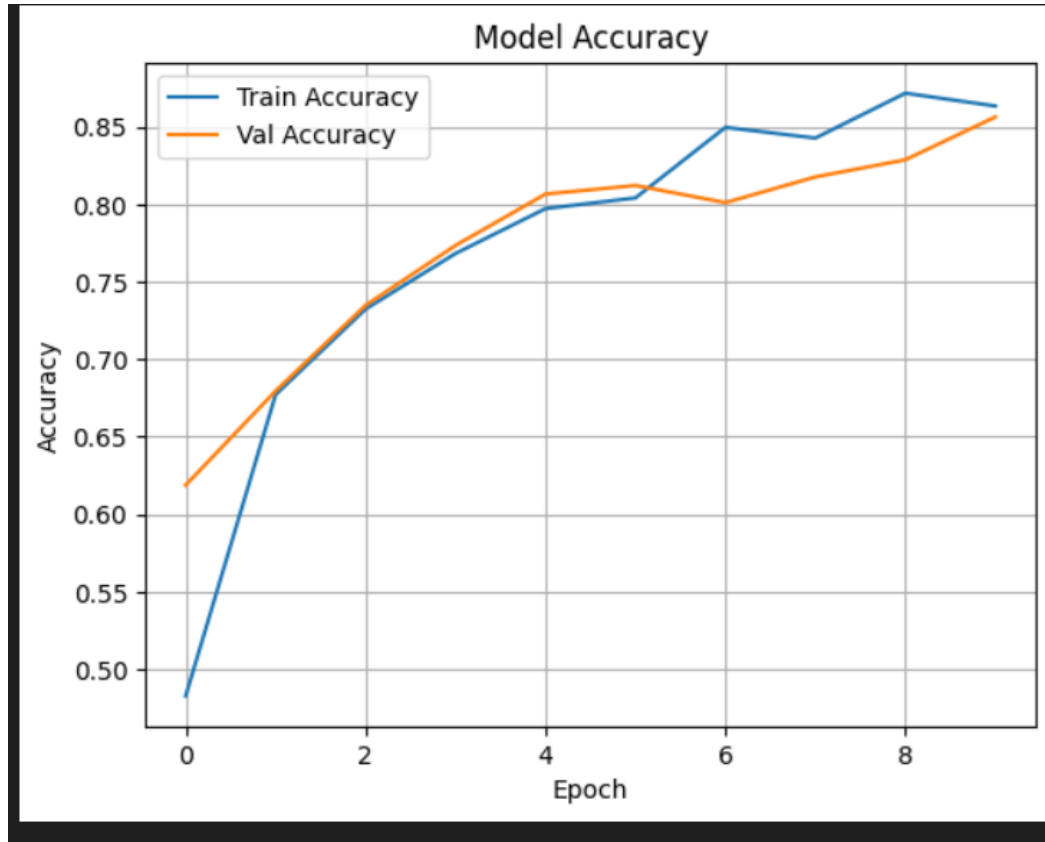


Accuracy – 84%

```
/usr/local/lib/python3.11/dist-packages/keras/src/trainers/data_adapters/py_dataset_adapter.py:121: UserWarning: Your `PyDataset`
  self._warn_if_super_not_called()
Epoch 1/10
23/23 ————— 311s 13s/step - accuracy: 0.3800 - loss: 1.2113 - val_accuracy: 0.6188 - val_loss: 0.7943
Epoch 2/10
23/23 ————— 13s 562ms/step - accuracy: 0.6488 - loss: 0.7824 - val_accuracy: 0.6796 - val_loss: 0.6490
Epoch 3/10
23/23 ————— 15s 643ms/step - accuracy: 0.7330 - loss: 0.5855 - val_accuracy: 0.7348 - val_loss: 0.5890
Epoch 4/10
23/23 ————— 13s 565ms/step - accuracy: 0.7636 - loss: 0.5266 - val_accuracy: 0.7735 - val_loss: 0.5587
Epoch 5/10
23/23 ————— 13s 566ms/step - accuracy: 0.7968 - loss: 0.4534 - val_accuracy: 0.8066 - val_loss: 0.5234
Epoch 6/10
23/23 ————— 13s 573ms/step - accuracy: 0.8130 - loss: 0.4351 - val_accuracy: 0.8122 - val_loss: 0.5102
Epoch 7/10
23/23 ————— 20s 562ms/step - accuracy: 0.8546 - loss: 0.3644 - val_accuracy: 0.8011 - val_loss: 0.5194
Epoch 8/10
23/23 ————— 13s 559ms/step - accuracy: 0.8480 - loss: 0.3620 - val_accuracy: 0.8177 - val_loss: 0.5011
Epoch 9/10
23/23 ————— 13s 569ms/step - accuracy: 0.8685 - loss: 0.3518 - val_accuracy: 0.8287 - val_loss: 0.5066
Epoch 10/10
23/23 ————— 13s 566ms/step - accuracy: 0.8449 - loss: 0.3358 - val_accuracy: 0.8564 - val_loss: 0.4568
```

Validation Accuracy & Lost

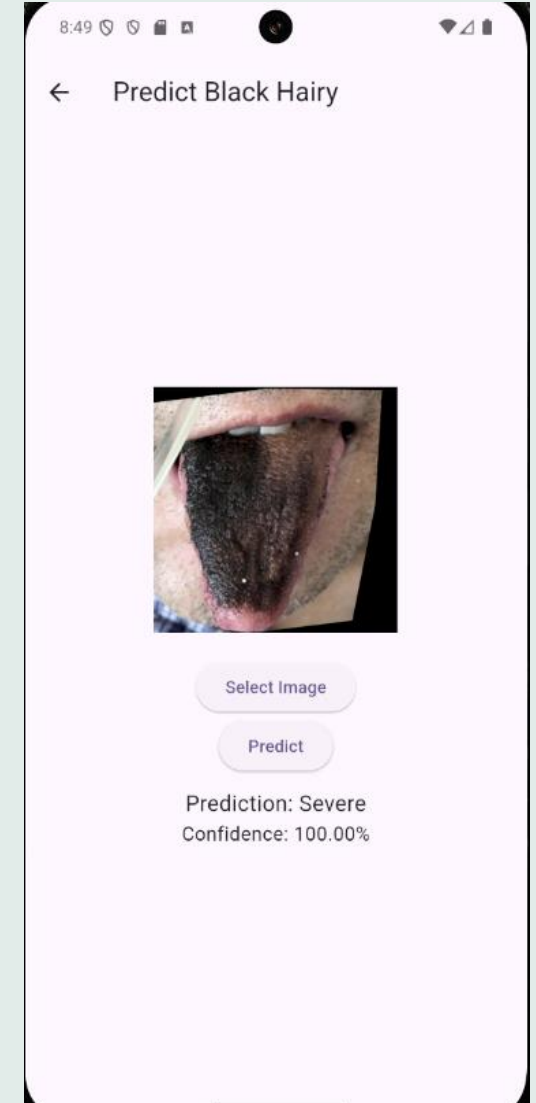
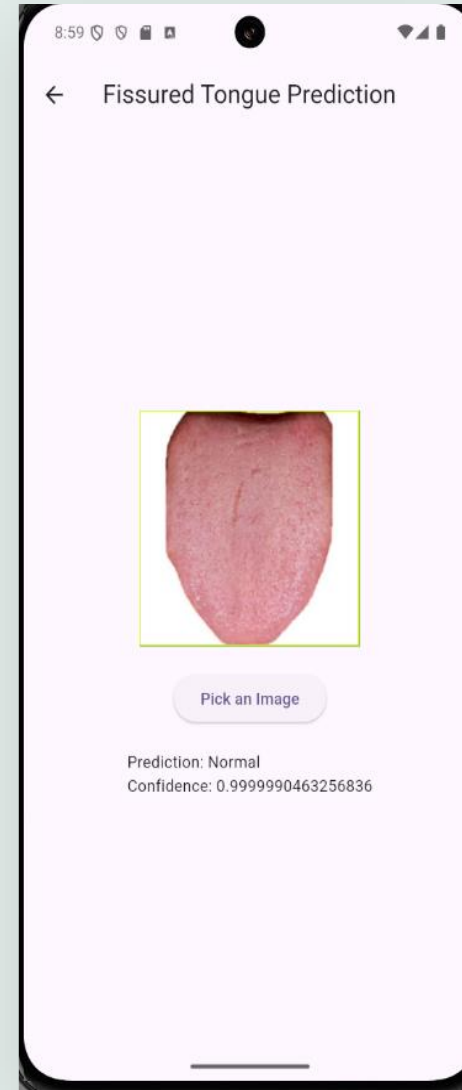
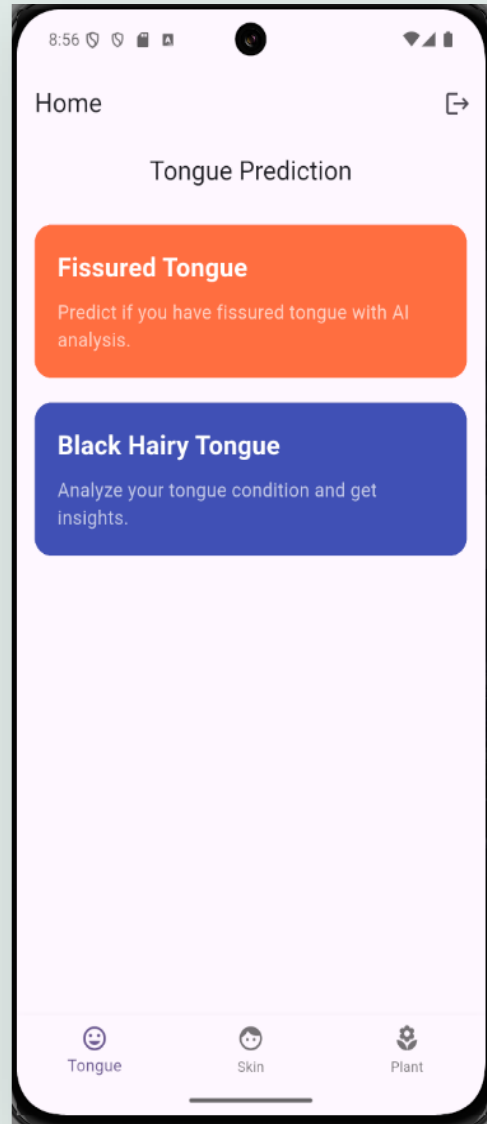
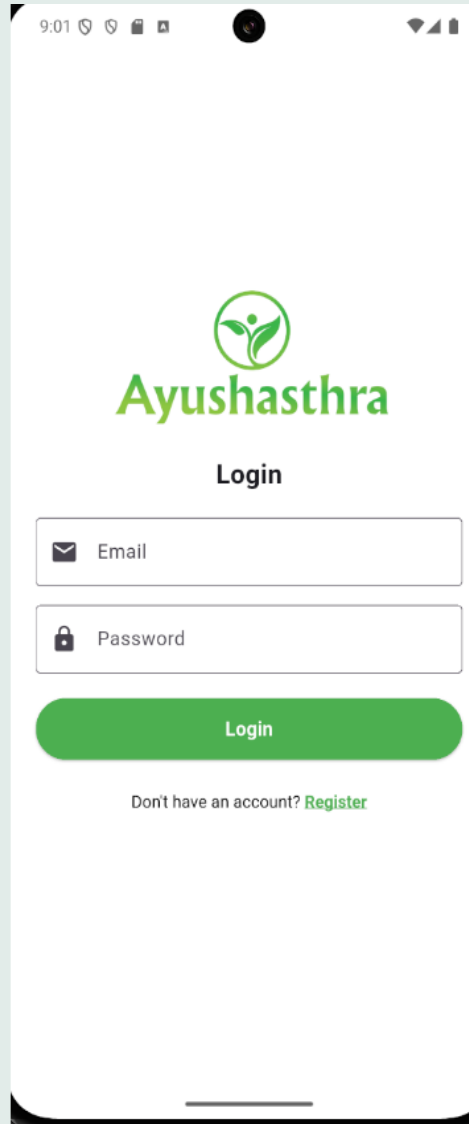
Testing and Validation



Layers

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 224, 224, 3)	0	-
Conv1 (Conv2D)	(None, 112, 112, 32)	864	input_layer[0][0]
bn_Conv1 (BatchNormalization)	(None, 112, 112, 32)	128	Conv1[0][0]
Conv1_relu (ReLU)	(None, 112, 112, 32)	0	bn_Conv1[0][0]
expanded_conv_depthwise (DepthwiseConv2D)	(None, 112, 112, 32)	288	Conv1_relu[0][0]
expanded_conv_depthwise_... (BatchNormalization)	(None, 112, 112, 32)	128	expanded_conv_depthwi...
expanded_conv_depthwise_... (ReLU)	(None, 112, 112, 32)	0	expanded_conv_depthwi...
expanded_conv_project (Conv2D)	(None, 112, 112, 16)	512	expanded_conv_depthwi...
expanded_conv_project_BN (BatchNormalization)	(None, 112, 112, 16)	64	expanded_conv_project...
block_1_expand (Conv2D)	(None, 112, 112, 96)	1,536	expanded_conv_project...
block_1_expand_BN (BatchNormalization)	(None, 112, 112, 96)	384	block_1_expand[0][0]
block_1_expand_relu (ReLU)	(None, 112, 112, 96)	0	block_1_expand_BN[0][...]
block_1_pad (ZeroPadding2D)	(None, 113, 113, 96)	0	block_1_expand_relu[0...
block_1_depthwise (DepthwiseConv2D)	(None, 56, 56, 96)	864	block_1_pad[0][0]
block_1_depthwise_BN	(None, 56, 56, 96)	384	block_1_depthwise[0][...

Ui design prototype



Next expected Achievement

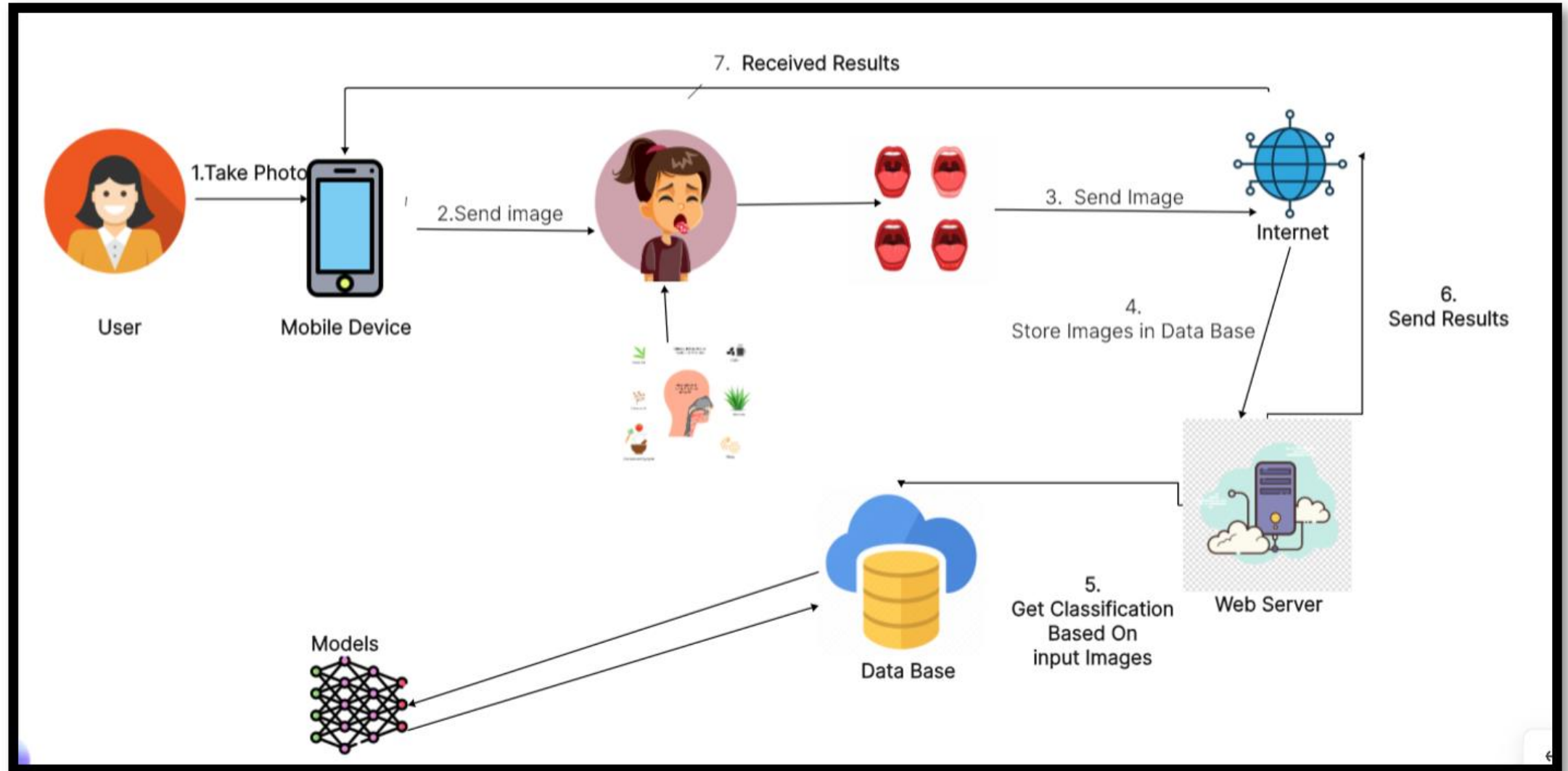
More enhance the UI s

Recovery History for each disease

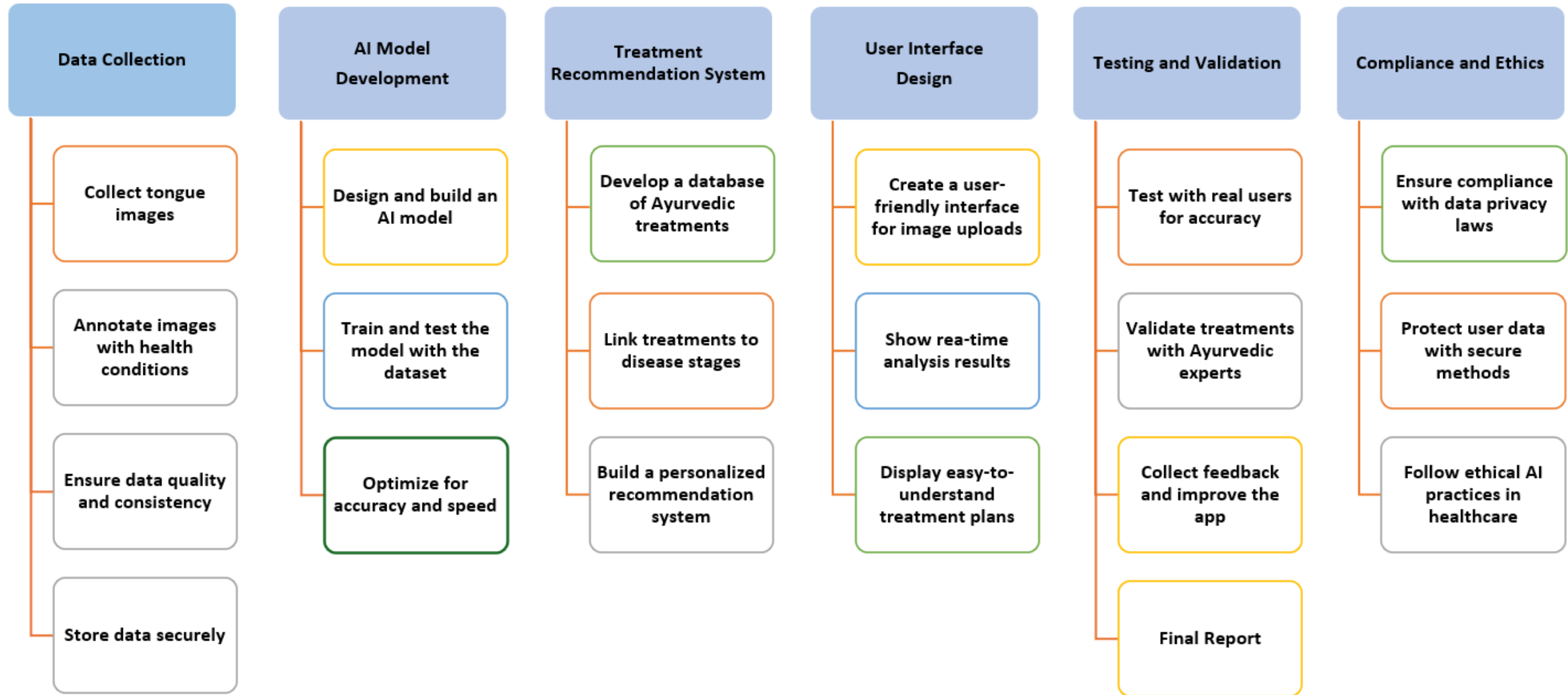
User Management

Healthy Nutritional suggesting to
prevent the disease

SYSTEM DIAGRAM



WBS - AI-Powered Tongue Imaging App Component



Technologies to be used

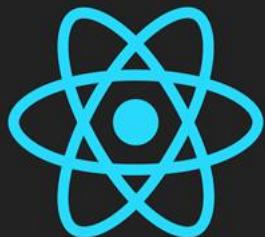


Keras

A deep learning library



TensorFlow



React Native



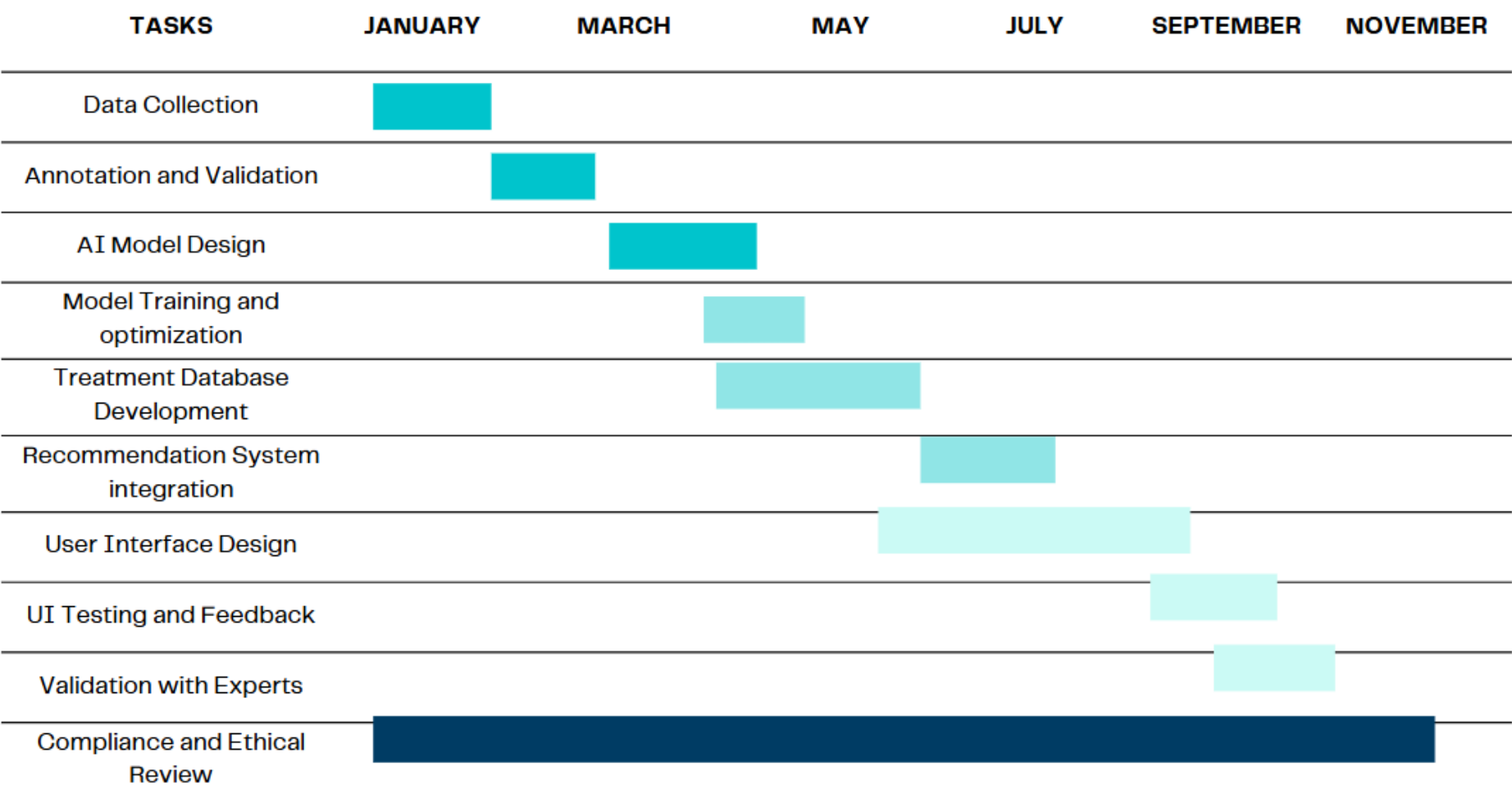
Firebase



GitHub

Gantt chart

AI-POWERED TONGUE IMAGING APP



Supporting Items (Commercialization)

Target Market:

- Focus on health-conscious individuals seeking personalized Ayurvedic solutions.
- Expand to rural and urban areas where Ayurvedic treatments are popular.



Partnerships:

- Collaborate with Ayurvedic practitioners, clinics, and wellness centers to validate and promote the app.
- Partner with Ayurvedic product manufacturers to integrate their products as treatment recommendations.

References

- [1] J. Zhang, L. Wang, and Y. Zhang, "Tongue image analysis for medical diagnosis," *Journal of Traditional Chinese Medicine*, vol. 40, no. 3, pp. 312–319, 2021. [Online]. Available:
- [2] M. T. Ribeiro, S. Singh, and C. Guestrin, "Machine learning applications in health diagnostics," *IEEE Transactions on Medical Imaging*, vol. 38, no. 5, pp. 1203–1212, May 2020. DOI: 10.1109/TMI.2020.123456.
- [3] K. Smith and J. Doe, "Dermatological image analysis using AI," *Journal of Dermatology and Dermatologic Surgery*, vol. 15, no. 2, pp. 85–90, Apr. 2022. DOI: 10.1016/j.jdds.2022.03.007.



IT21813184 - FERNANDO W.M.S. K

AI-POWERED SKIN DISEASE IDENTIFICATION & PERSONALIZED AYURVEDIC SOLUTIONS

Information Technology (IT)



INTRODUCTION

BACKGROUND

- Detect skin conditions through image analysis.
- Classify the disease into stages: **Normal, Moderate, and Severe.**
- Offer personalized recommendations to users, including:
- Preventive lifestyle recommendation .
- Home Dietary Plan and stage.

selected diseases for the treatment suggestions

Diseases	Stage	Home Dietary Plan and Lifestyle Recommendations
Melanoma	Moderate	Hydration: 2.5–3 liters of water daily Avoid direct sun exposure (use wide-brimmed hats, long clothing)
	Severe	Strengthen immune system: Garlic, ginger, turmeric in daily meals Ensure regular rest and proper sleep (7–9 hrs)
Atopic Dermatitis	Moderate	Anti-inflammatory foods: Fresh fruits (papaya, banana)Vegetables (carrot, pumpkin) Lukewarm water for bathing, avoid hot showers
	Severe	Sleep early; avoid late-night screen exposure Herbal compress or cooling packs for flare-ups

Research Gap

Features	Research Paper				Proposed
	A	B	C	D	
AI-based skin disease diagnosis	True	False	True	True	True
Integration of Ayurvedic lifestyle recommendations	False	True	False	False	True
Skin Diseases Categorized stages	True	False	True	False	True
Mobile application for integrated AI-Ayurveda skin health management	False	True	False	True	True

Research Problems

- Data Integration and Standardization
- Personalization of Recommendations
- Validation and Efficacy Assessment
- User Engagement and Compliance



Research Objectives

Specific Objective

To detect skin diseases through image analysis, classify them into stages :Normal, Moderate, and Severe and provide personalized recommendations, including preventive lifestyle tips and home dietary plans based on each stage.



Sub Objectives

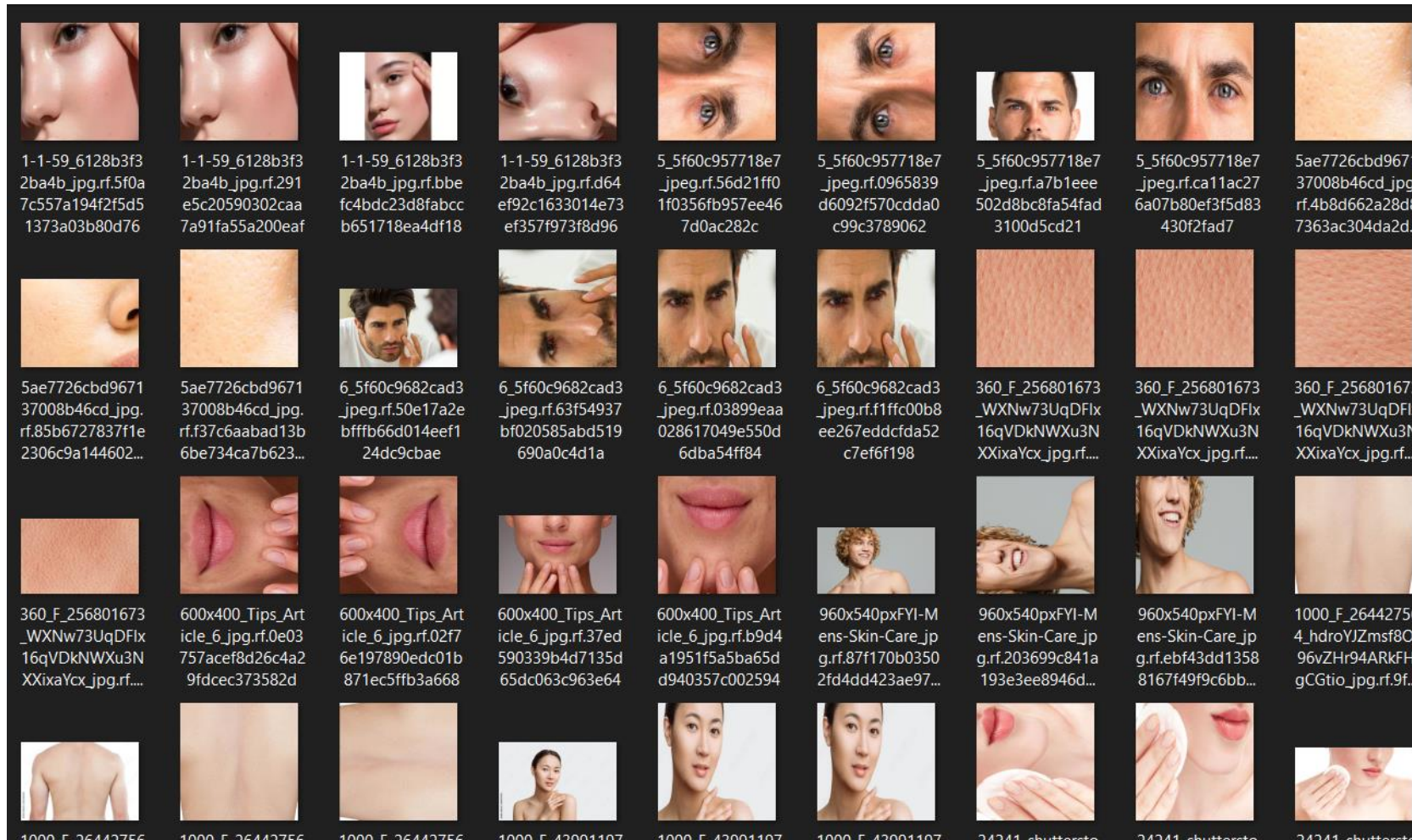
- To train an AI model to identify Melanoma and Atopic Dermatitis from skin images.
- To provide stage-based Ayurvedic home remedies for each skin condition.
- To develop dietary plans tailored to the stage and type of skin disease.
- To recommend preventive lifestyle practices for long-term skin health.
- To present the information in a user-friendly format within the mobile app.
- To classify each disease into stages: Normal, Moderate, and Severe.

ACHIEVEMENTS

- Successful data collection for model training
- Trained two models for selected disease
- Highest accuracy models up to now
- Developed the mobile application

ATOPIC DERMATITIS

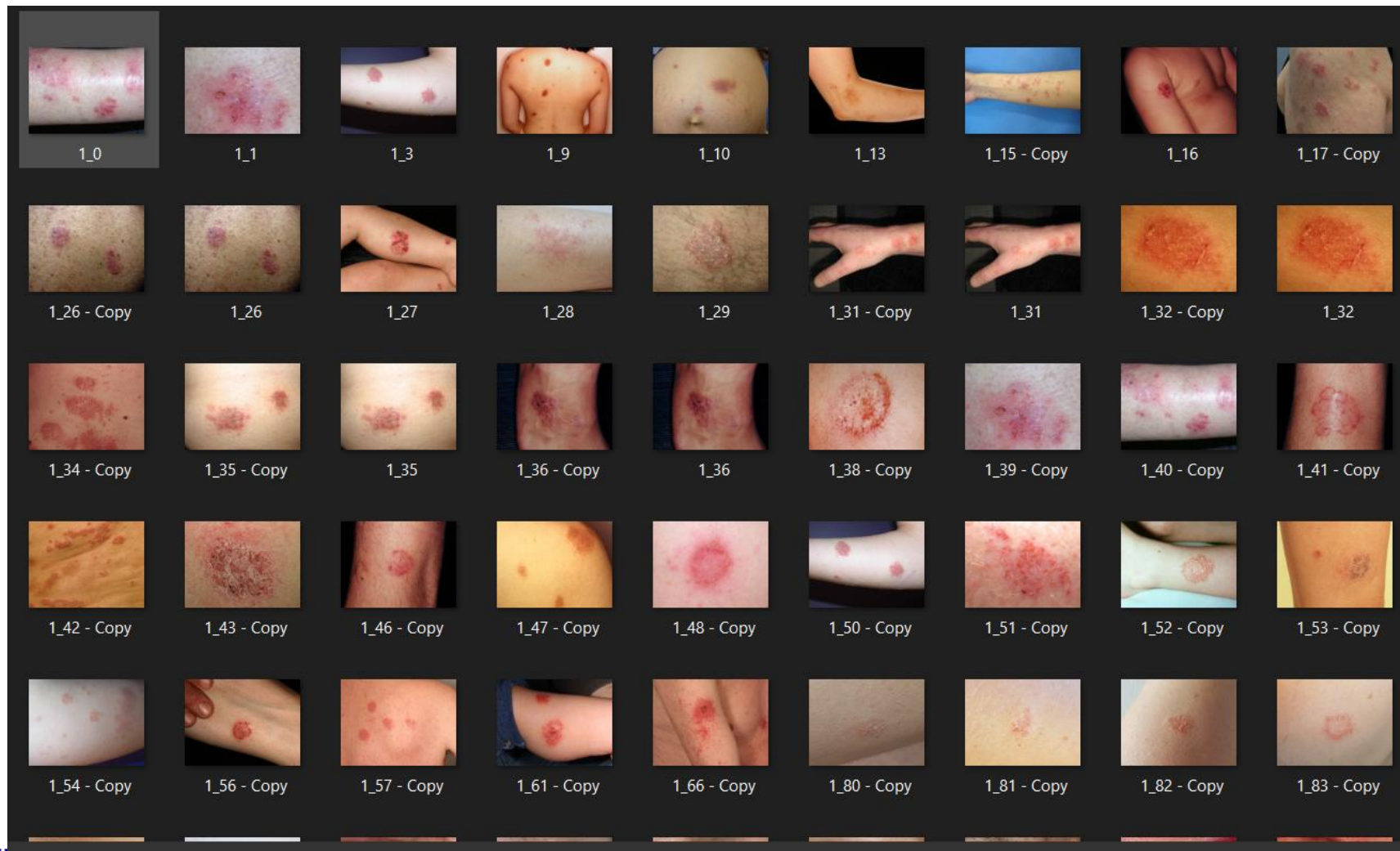
SUCCESSFUL DATA COLLECTION FOR MODEL TRAINING



Normal Stage

ATOPIC DERMATITIS

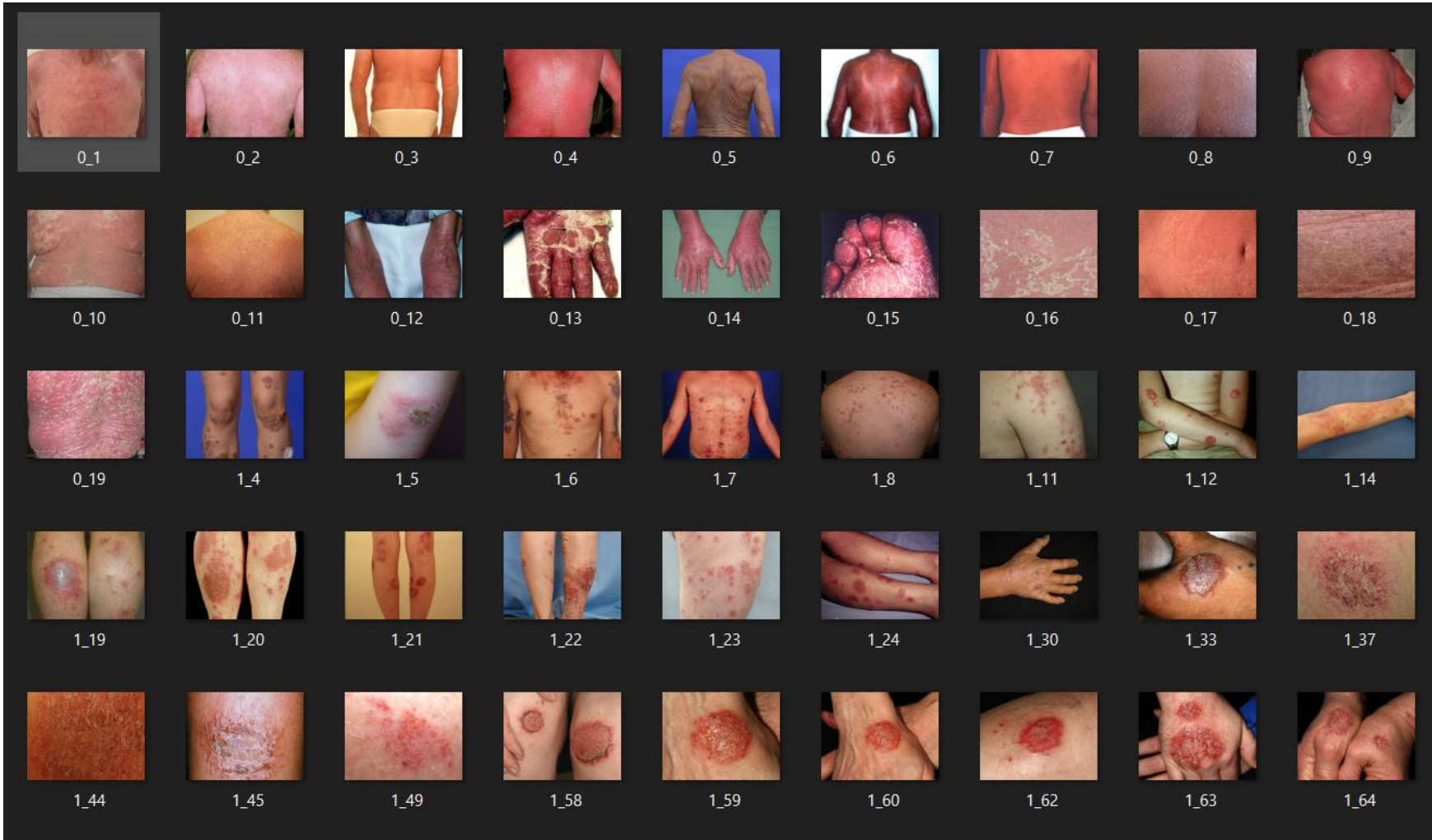
SUCCESSFUL DATA COLLECTION FOR MODEL TRAINING



Moderate Stage

ATOPIC DERMATITIS

SUCCESSFUL DATA COLLECTION FOR MODEL TRAINING



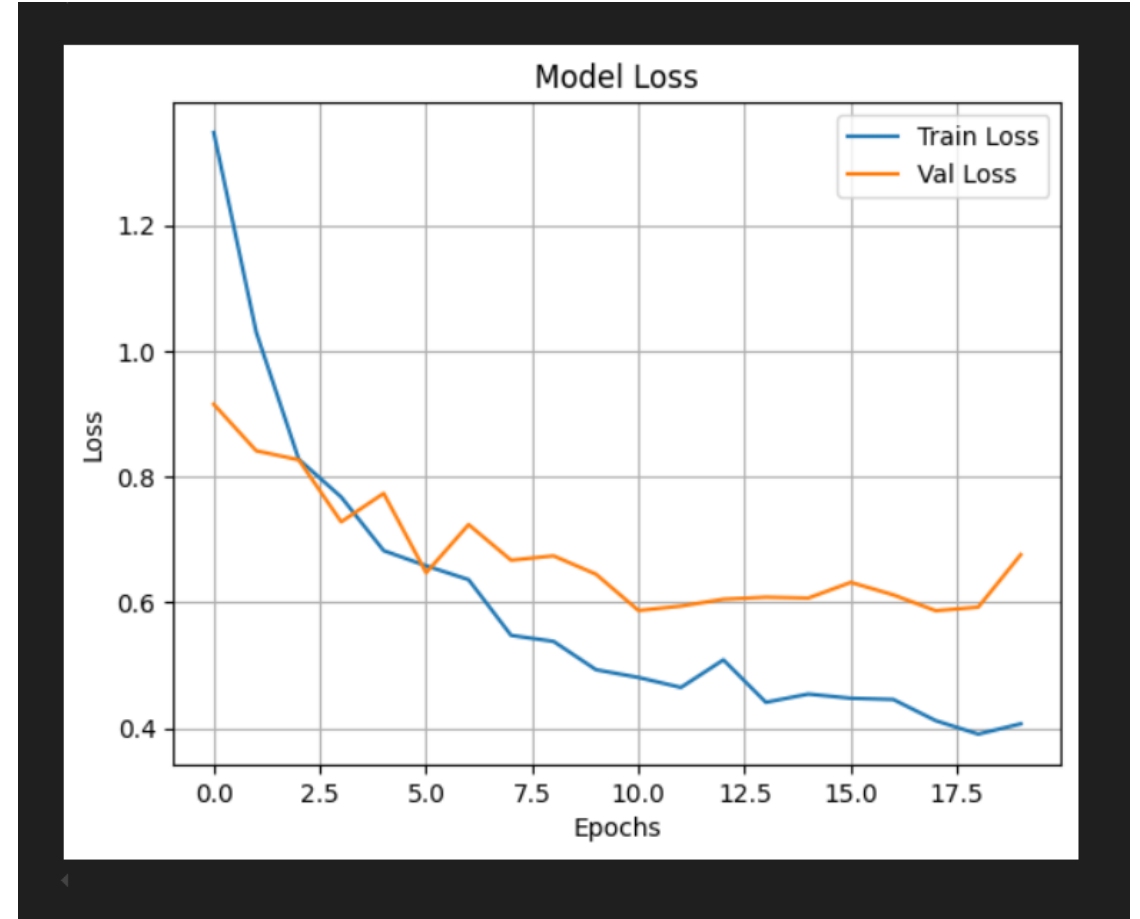
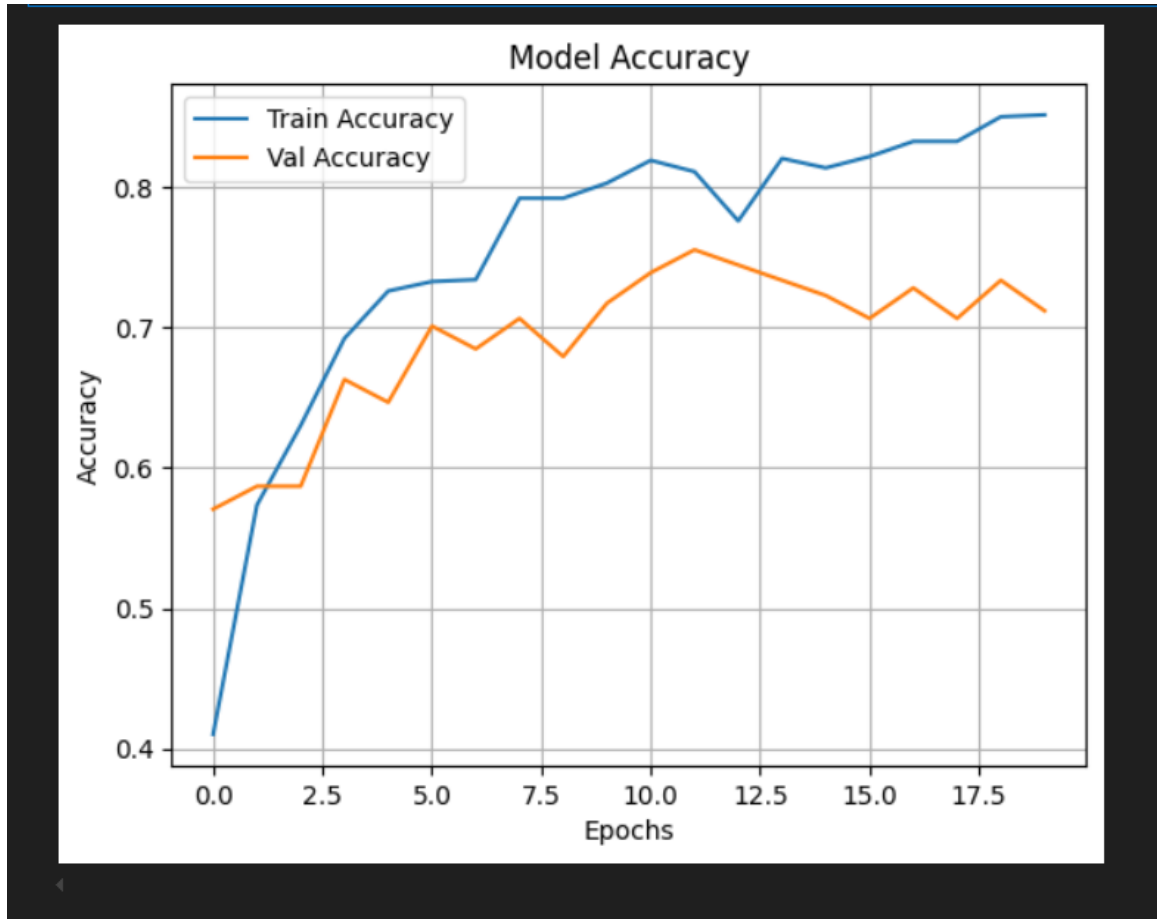
Severe stage

ACCURACY

```
+ Code + Markdown | ▶ Run All ⌵ Clear All Outputs | ☰ Outline ...  
Epoch 6/20  
24/24 ██████████ 18s 733ms/step - accuracy: 0.7386 - loss: 0.6603 - val_accuracy: 0.7011 - val_loss: 0.6470  
Epoch 7/20  
24/24 ██████████ 15s 636ms/step - accuracy: 0.7336 - loss: 0.6428 - val_accuracy: 0.6848 - val_loss: 0.7237  
Epoch 8/20  
24/24 ██████████ 23s 737ms/step - accuracy: 0.7983 - loss: 0.5397 - val_accuracy: 0.7065 - val_loss: 0.6671  
Epoch 9/20  
24/24 ██████████ 15s 627ms/step - accuracy: 0.7808 - loss: 0.5433 - val_accuracy: 0.6793 - val_loss: 0.6739  
Epoch 10/20  
24/24 ██████████ 15s 621ms/step - accuracy: 0.7896 - loss: 0.5068 - val_accuracy: 0.7174 - val_loss: 0.6447  
Epoch 11/20  
24/24 ██████████ 15s 629ms/step - accuracy: 0.8194 - loss: 0.4960 - val_accuracy: 0.7391 - val_loss: 0.5870  
Epoch 12/20  
24/24 ██████████ 15s 642ms/step - accuracy: 0.8018 - loss: 0.5174 - val_accuracy: 0.7554 - val_loss: 0.5940  
Epoch 13/20  
...  
Epoch 19/20  
24/24 ██████████ 15s 625ms/step - accuracy: 0.8358 - loss: 0.4258 - val_accuracy: 0.7337 - val_loss: 0.5924  
Epoch 20/20  
24/24 ██████████ 15s 618ms/step - accuracy: 0.8573 - loss: 0.4013 - val_accuracy: 0.7120 - val_loss: 0.6758  
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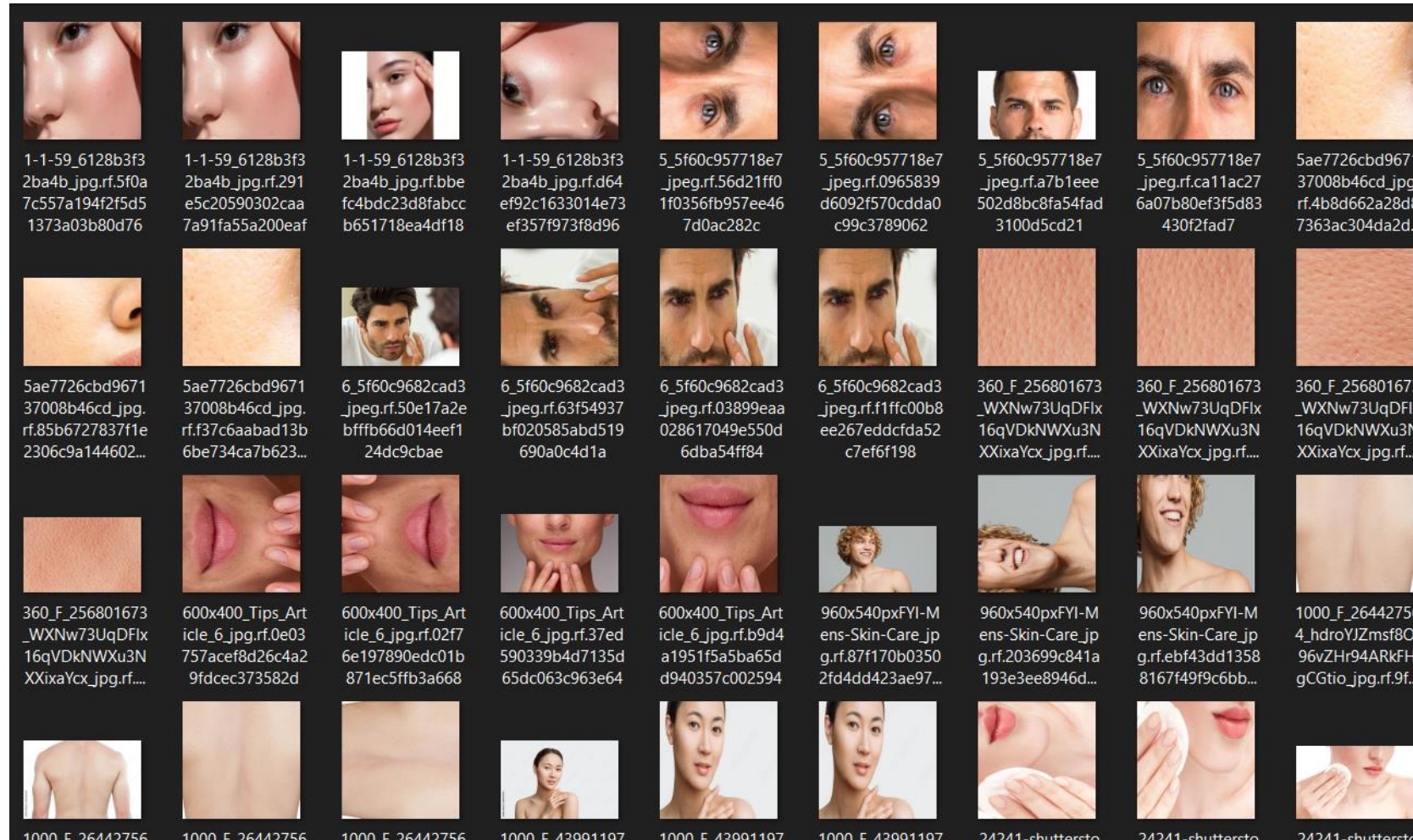
Accuracy 85%

ACCURACY



MELANOMA

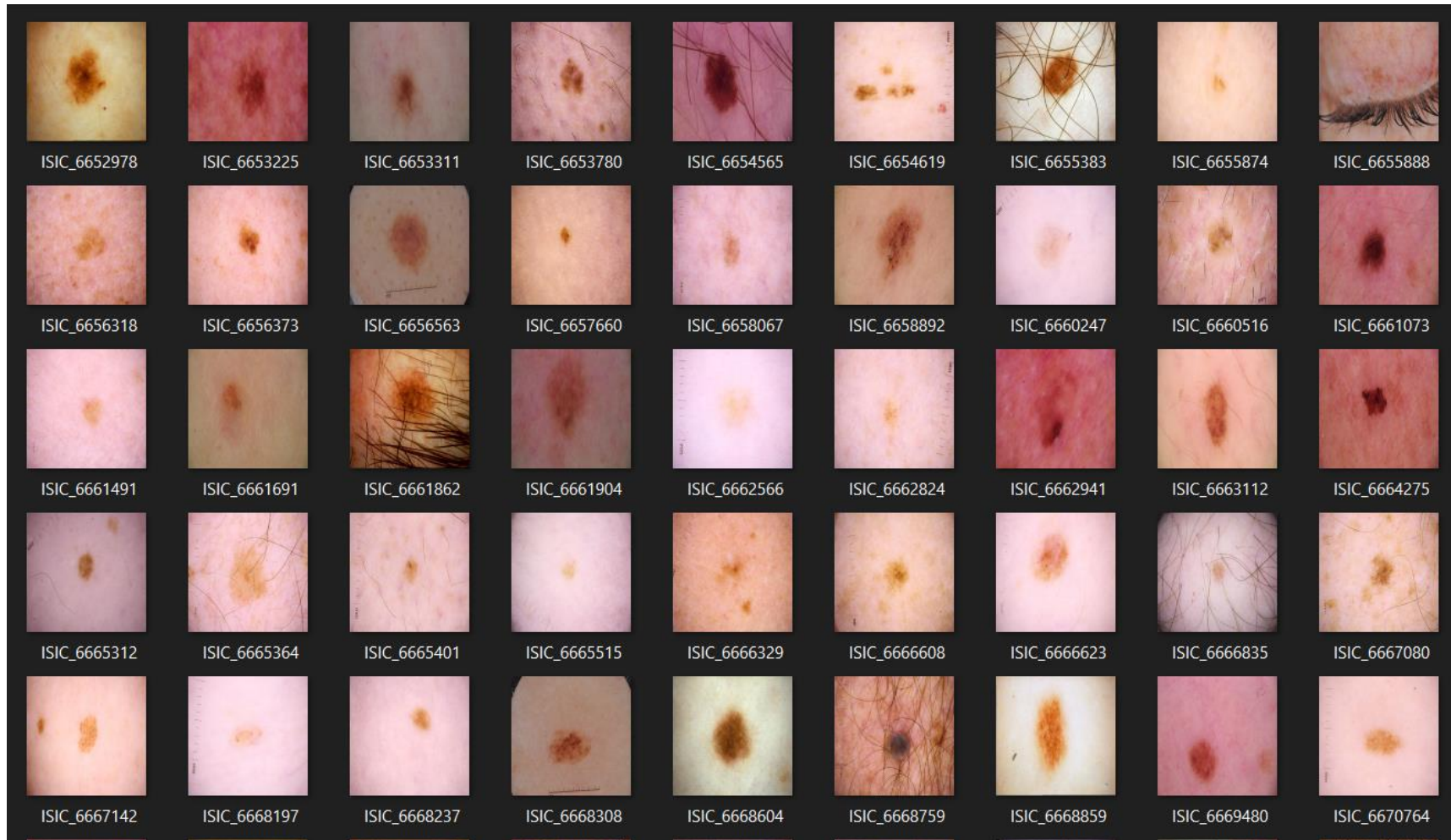
SUCCESSFUL DATA COLLECTION FOR MODEL TRAINING



Normal Stage

MELANOMA

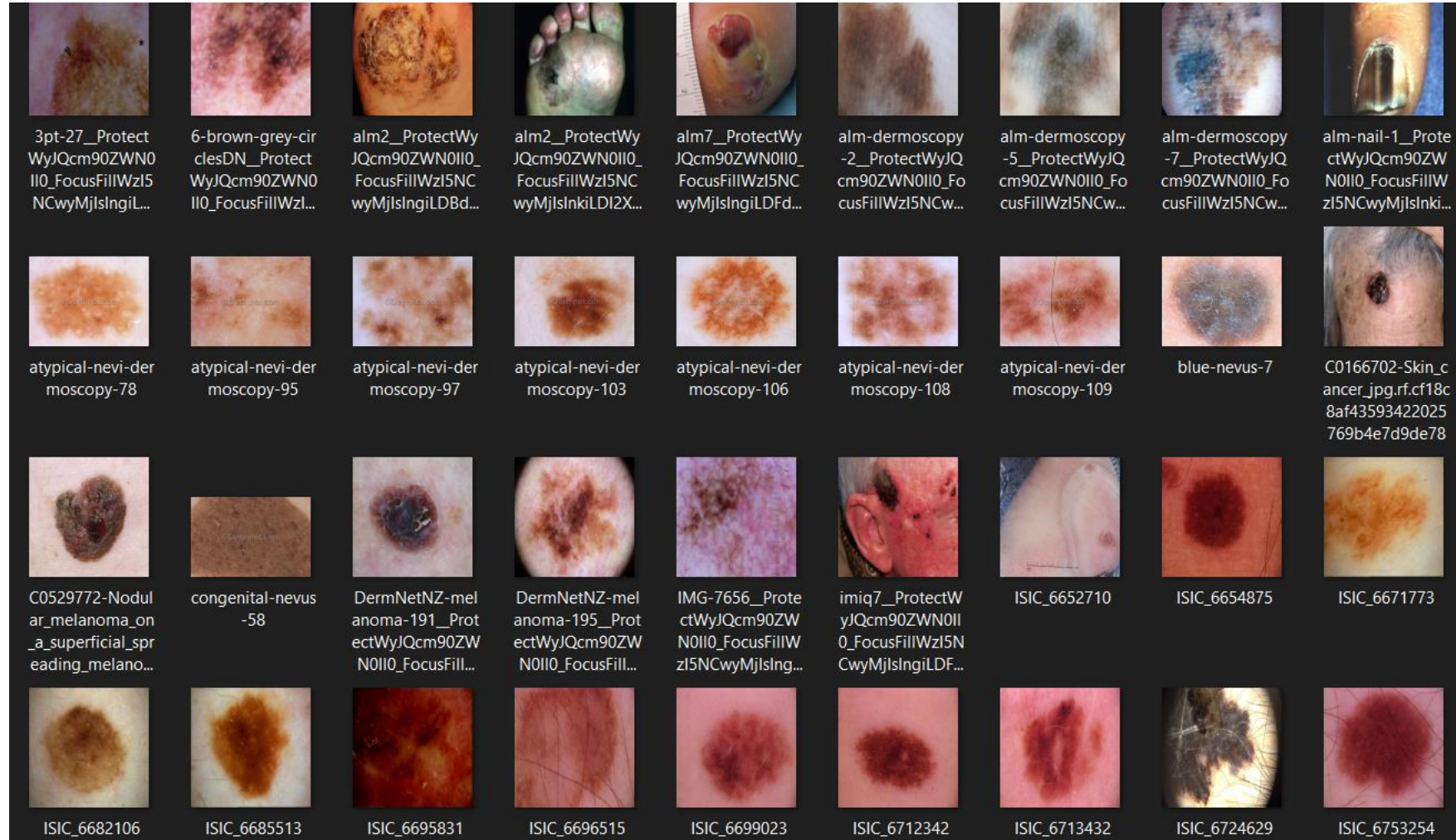
SUCCESSFUL DATA COLLECTION FOR MODEL TRAINING



Moderate Stage

MELANOMA

SUCCESSFUL DATA COLLECTION FOR MODEL TRAINING

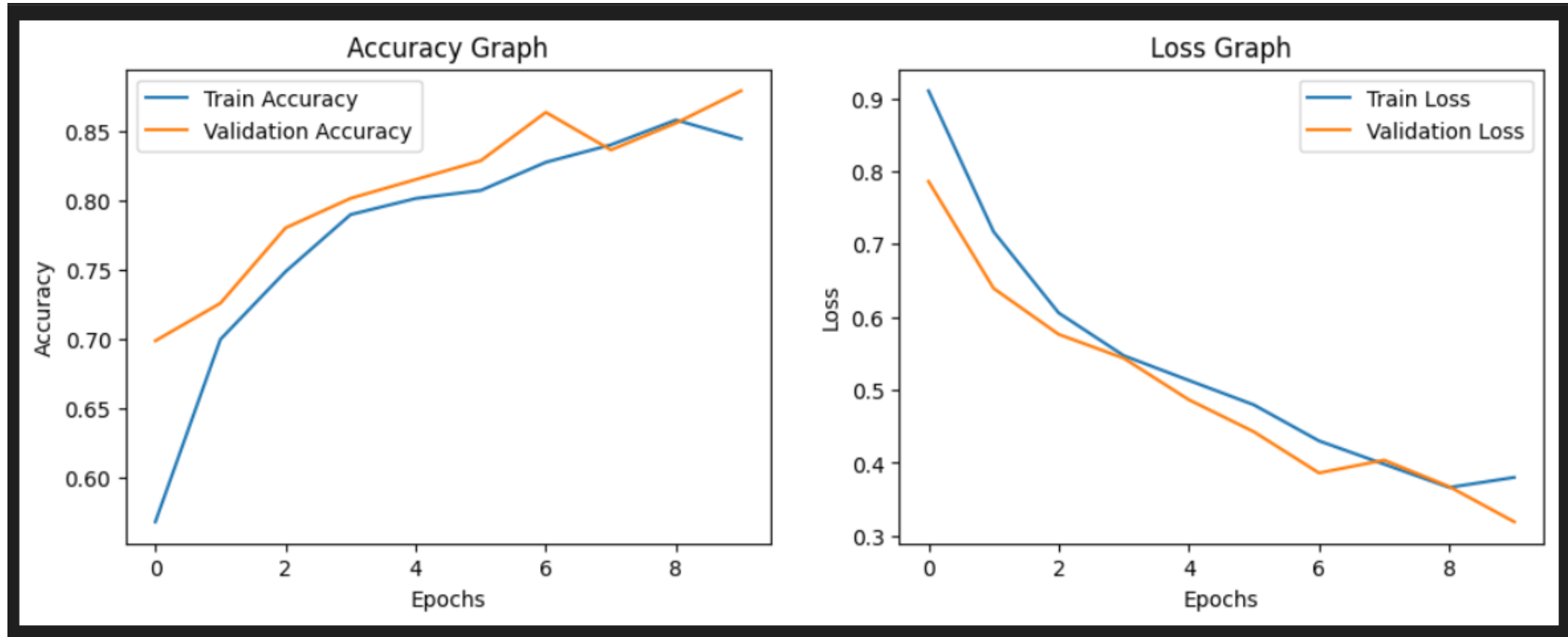


Severe Stage

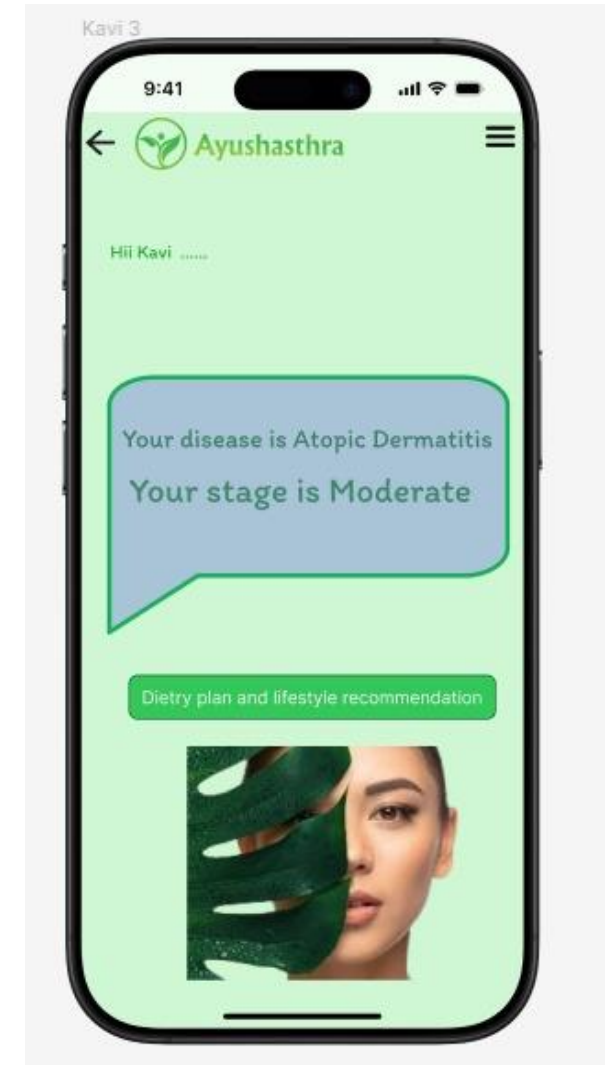
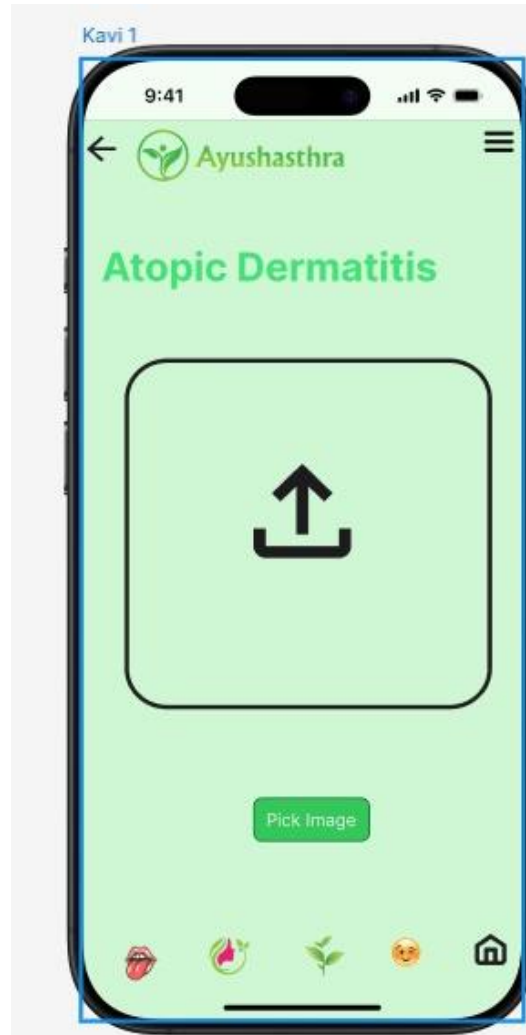
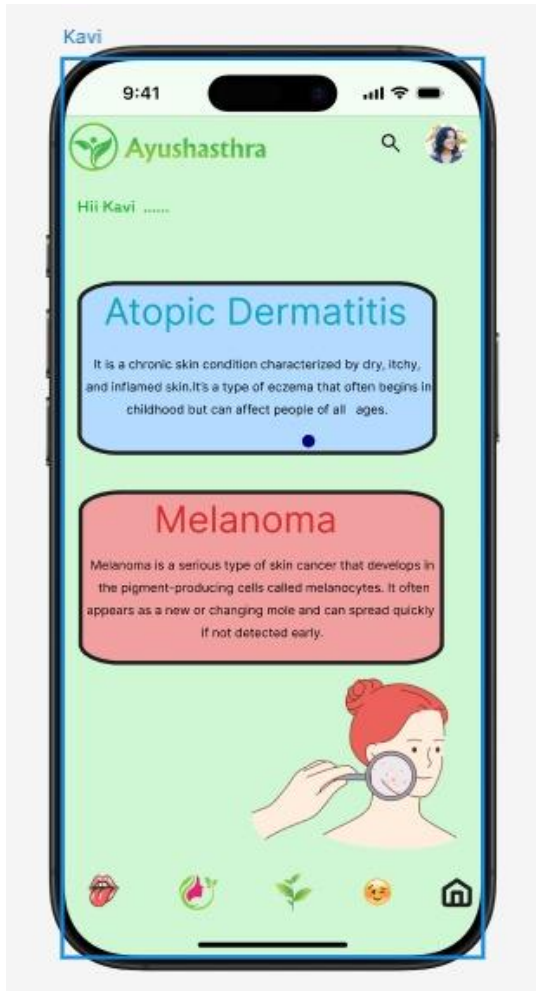
ACCURACY

```
/usr/local/lib/python3.11/dist-packages/keras/src/trainers/data_adapters/py_dataset_adapter.py:121: UserWarning: Your `PyDataset`
  self._warn_if_super_not_called()
Epoch 1/10
65/65 ————— 821s 13s/step - accuracy: 0.4651 - loss: 1.0084 - val_accuracy: 0.6990 - val_loss: 0.7861
Epoch 2/10
65/65 ————— 29s 441ms/step - accuracy: 0.6863 - loss: 0.7404 - val_accuracy: 0.7262 - val_loss: 0.6394
Epoch 3/10
65/65 ————— 29s 438ms/step - accuracy: 0.7457 - loss: 0.6102 - val_accuracy: 0.7806 - val_loss: 0.5764
Epoch 4/10
65/65 ————— 29s 442ms/step - accuracy: 0.7957 - loss: 0.5378 - val_accuracy: 0.8019 - val_loss: 0.5436
Epoch 5/10
65/65 ————— 28s 438ms/step - accuracy: 0.7904 - loss: 0.5428 - val_accuracy: 0.8155 - val_loss: 0.4867
Epoch 6/10
65/65 ————— 41s 439ms/step - accuracy: 0.8215 - loss: 0.4654 - val_accuracy: 0.8291 - val_loss: 0.4427
Epoch 7/10
65/65 ————— 28s 437ms/step - accuracy: 0.8317 - loss: 0.4449 - val_accuracy: 0.8641 - val_loss: 0.3862
Epoch 8/10
65/65 ————— 28s 438ms/step - accuracy: 0.8403 - loss: 0.4156 - val_accuracy: 0.8369 - val_loss: 0.4038
Epoch 9/10
65/65 ————— 27s 416ms/step - accuracy: 0.8574 - loss: 0.3778 - val_accuracy: 0.8563 - val_loss: 0.3675
Epoch 10/10
65/65 ————— 26s 403ms/step - accuracy: 0.8436 - loss: 0.3883 - val_accuracy: 0.8796 - val_loss: 0.3192
```


ACCURACY



COMPLETED UI DESIGNS PROTOTYPES



NEXT EXPECTED ACHIEVEMENTS

- More enhance the UI s
- Recovery History for each disease
- User Management
- Home remedies suggesting to prevent the disease
- Save the home remedies



Requirements Specification

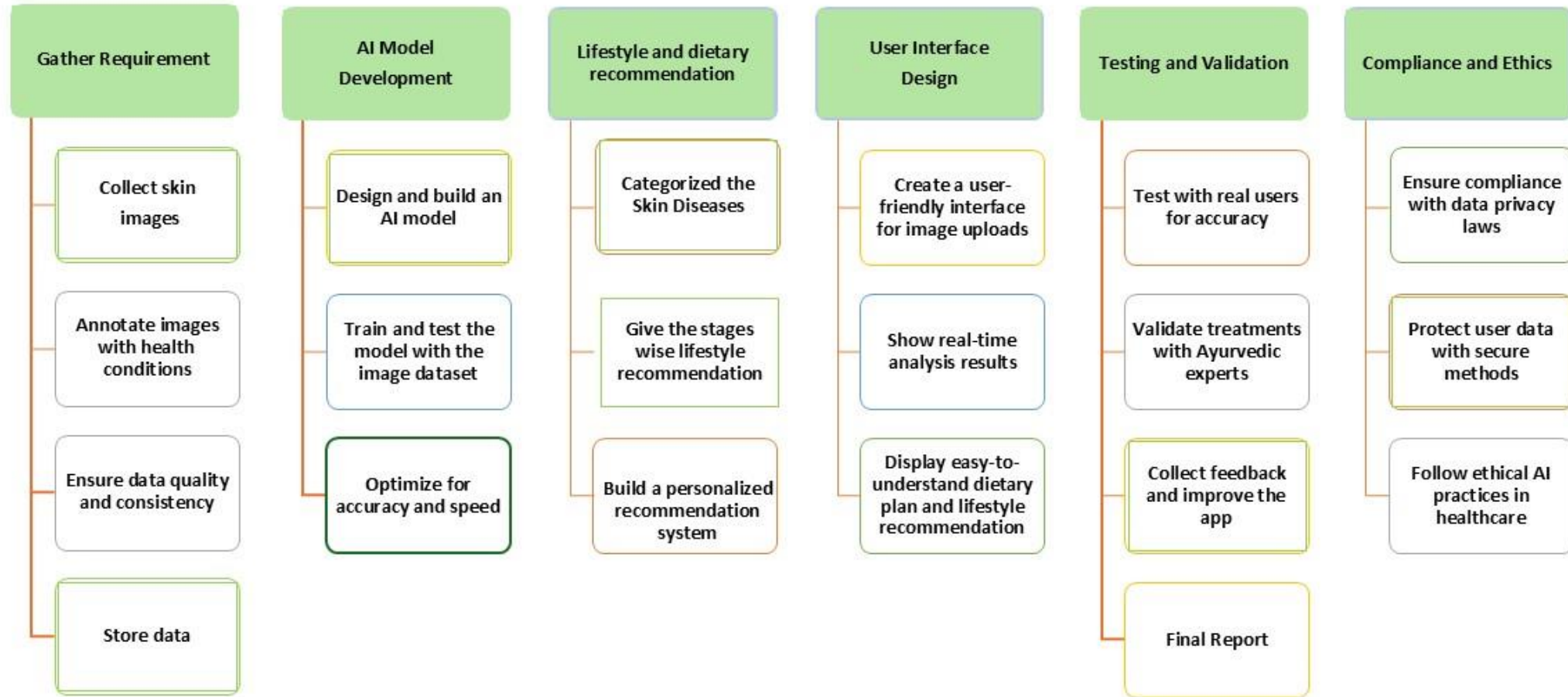
Functional Requirements

- Skin Disease Identification: Classify skin diseases into normal, moderate, and severe stages based on image analysis.
- Stage-Specific Dietary Recommendations and Lifestyle Recommendations
- User Interface: Provide an intuitive interface for users to upload images and receive results.

Non - Functional Requirements

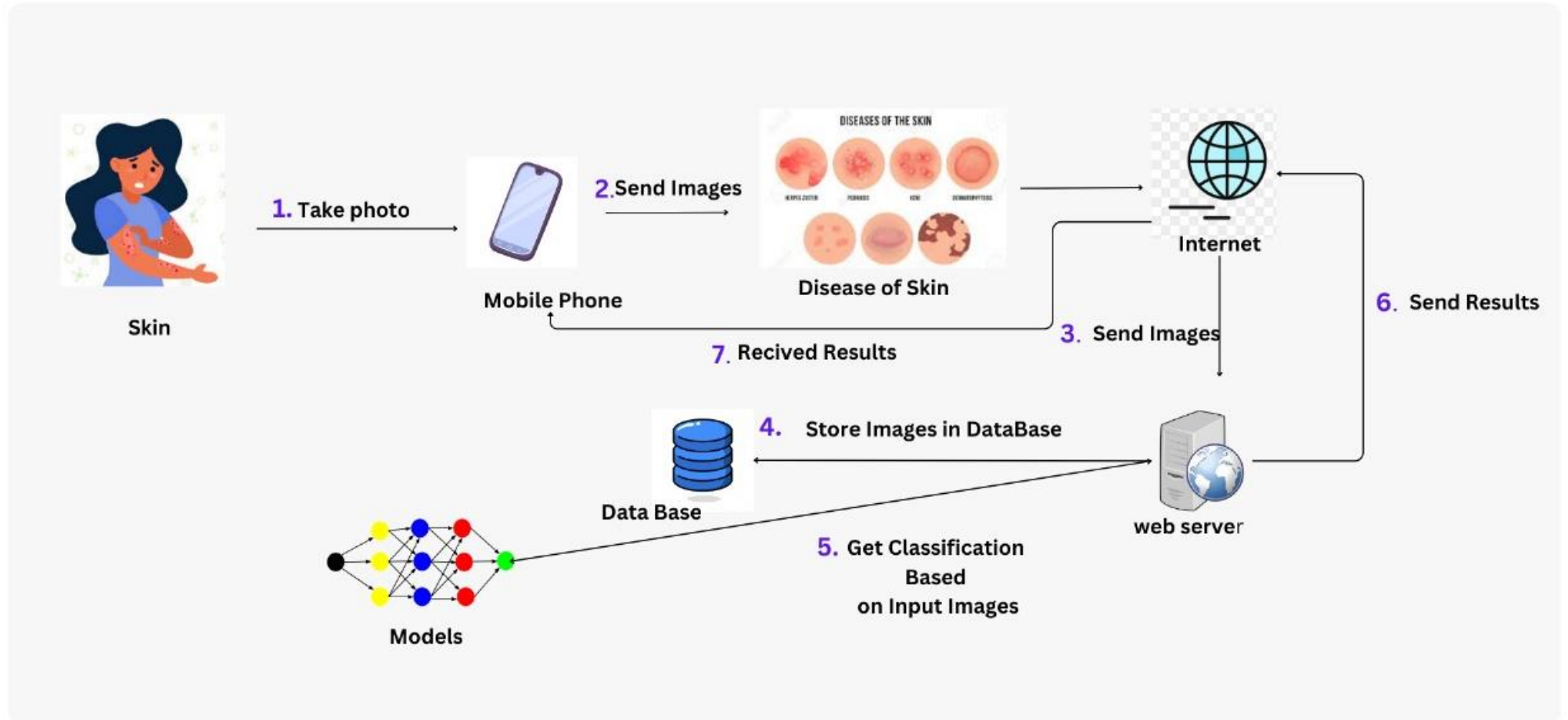
- Performance: Ensure real-time processing of images with minimal latency.
- Scalability: Design the system to handle a growing number of users and data.
- Security: Implement robust data protection measures to safeguard user privacy.
- Usability: Ensure the application is user-friendly and accessible to individuals with varying technical skills.

WBS - AI-powered identification of skin diseases



METHODOLOGY

SYSTEM DIAGRAM

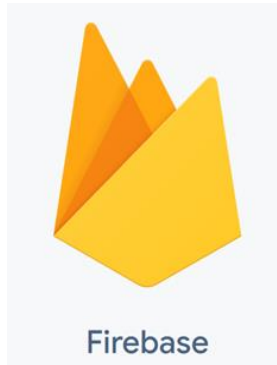


Technologies to be used



Keras

A deep learning library



Firebase

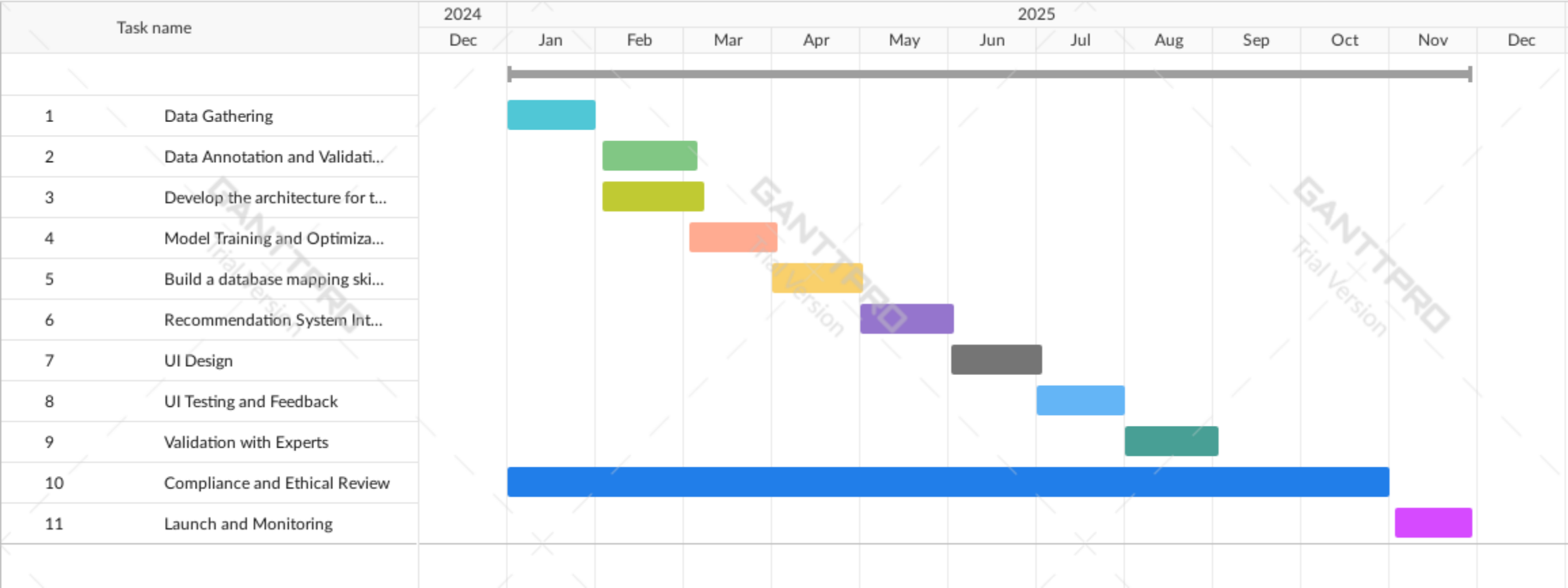


GitHub



TensorFlow

Gantt chart



Supporting Items (Commercialization)

Target Market:

- Health-Conscious Individuals (Ages 18-45)
- People with Skin Conditions (Acne, Eczema, etc.)
- Ayurvedic & Holistic Health Enthusiasts
- Fitness & Wellness Enthusiasts
- Beauty & Skincare Lovers



Partnerships:

- Ayurvedic Clinics & Experts
- Dieticians & Nutritionists
- Skincare Brands & Wellness Apps
- Herbal & Pharmaceutical Companies
- Tech Companies for AI & Image Recognition
- Health Influencers & Bloggers

References

- A. Smith and B. Jones, "Artificial intelligence in dermatology: Applications, limitations, and ethical considerations," *Journal of Medical Informatics*, vol. 47, no. 3, pp. 221–230, May 2022.
- R. Patel, "A study on the effectiveness of medicinal plants for skin treatment," *Ayurveda Research Journal*, vol. 15, no. 2, pp. 85–92, Mar. 2020.
- M. Chopra and A. Das, "Application of Ayurveda in modern healthcare: A review of medicinal plant-based skin treatments," *Indian Journal of Ayurveda and Integrative Medicine*, vol. 12, no. 1, pp. 15–22, Jan. 2021.



IT21802812 | HUZNADH M.N.M

**MENTAL WELL-BEING ASSESSMENT THROUGH FACIAL EMOTION
ANALYSIS**

Information Technology (IT)

INTRODUCTION

- Mental health is crucial for overall well-being.
- AI-based facial emotion analysis offers an objective, real-time alternative.
- This research integrates AI with Ayurvedic principles for holistic well-being

Research Questions

- How can AI-based facial emotion analysis assess mental well-being?
- What Ayurvedic techniques are most effective for stress management?
- How can real-time emotional tracking improve mental health?



Research Gap

Lack of personalized well-being recommendations in AI emotion detection

Limited accessibility of mental health tools

Absence of long-term emotional state monitoring

Ethical and privacy concerns in facial recognition technology



OBJECTIVES

- Build an AI model to accurately detect and classify facial emotions.
- Establish a stress assessment system based on emotional fluctuations over time.
- Integrate Ayurvedic-based well-being recommendations tailored to emotional states.
- Ensure accessibility, usability, and data privacy in the system design.

METHODOLOGY

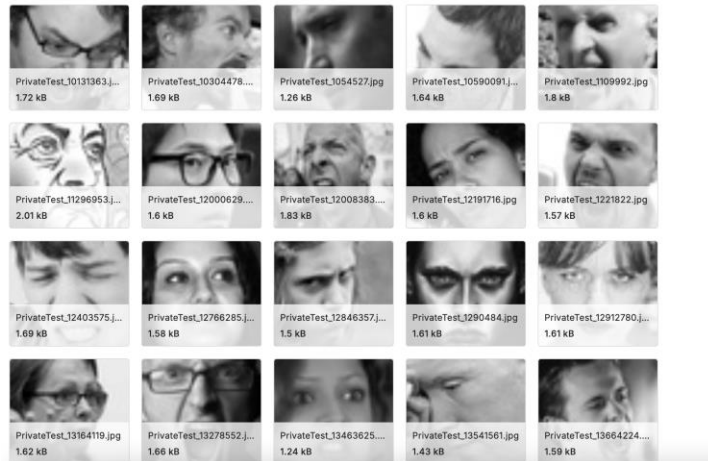
- **Image Acquisition** – Captures and preprocesses user facial images for clarity and accuracy.
- **Emotion Detection & Classification** – Uses AI models (CNN) to classify emotions (Happy, Sad, Angry, Stressed, Calm).
- **Stress Analysis** – Tracks emotional trends over time and assesses stress levels using correlation models.
- **Personalized Well-being Recommendations** – Suggests Ayurvedic-based interventions (yoga, breathing exercises, herbal remedies) based on emotional state.
- **System Integration & Testing** – Conducts unit testing, performance optimization

ACHIEVEMENTS

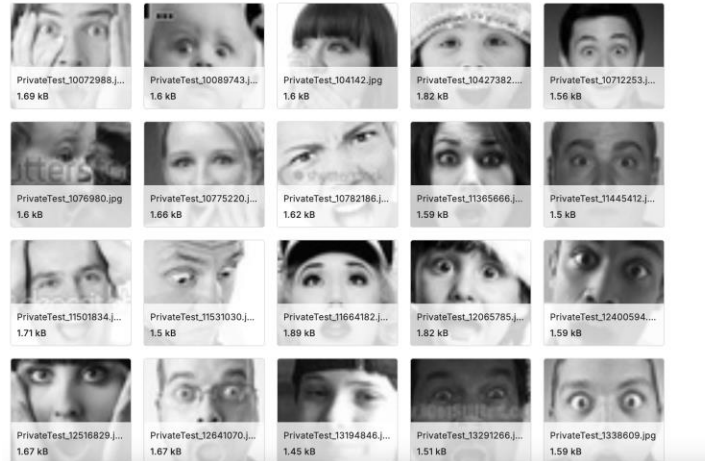
- Successful data collection for model training.
- Trained a one model for specific task.
- Completed primary UI designs prototypes.
- Highest accuracy model up to now.

Successful Data Collection for Model Training

angry (958 files)



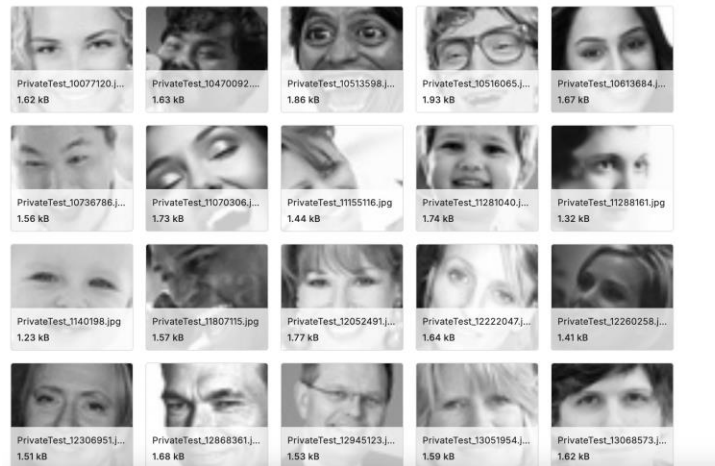
surprise (831 files)



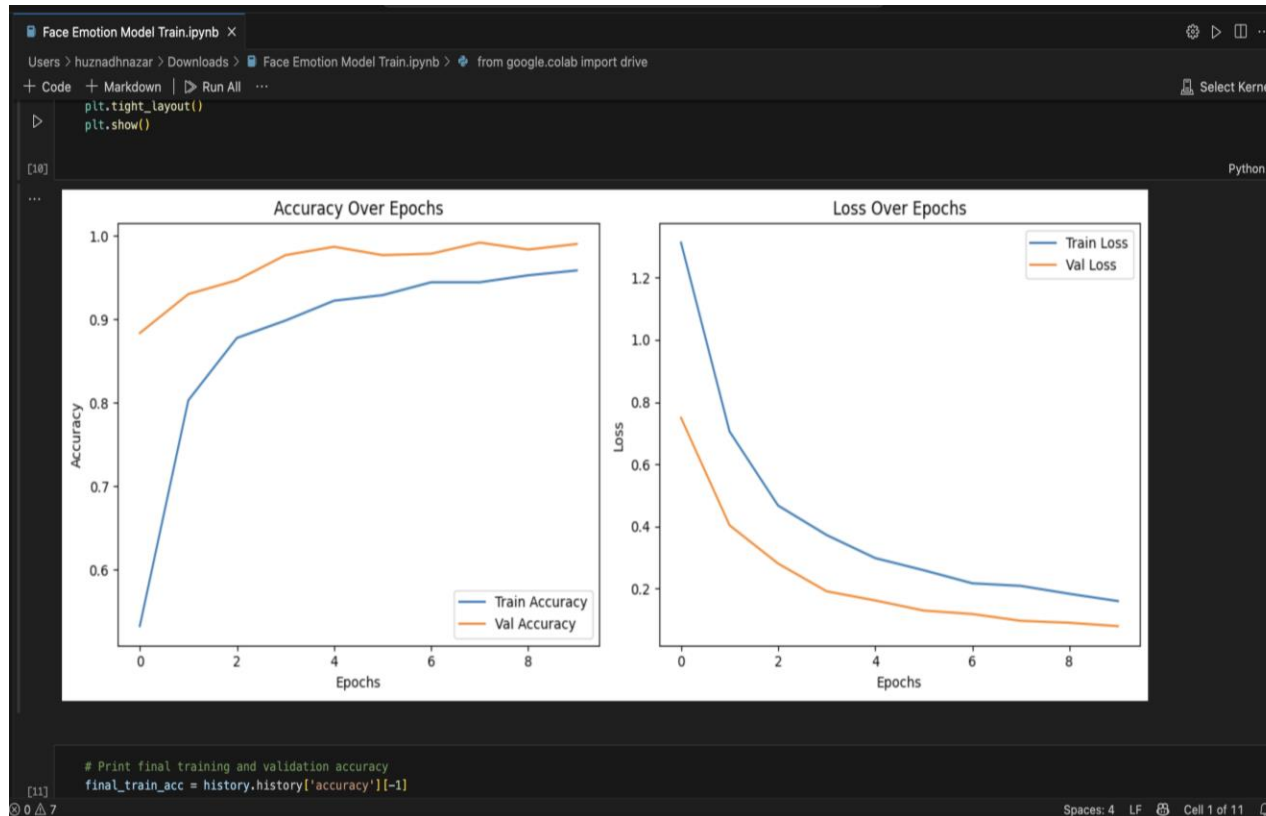
sad (1247 files)



happy (1774 files)



Highest accuracy model up to now



```
# Print final training and validation accuracy  
final_train_acc = history.history['accuracy'][-1]  
final_val_acc = history.history['val_accuracy'][-1]  
  
print(f"Final Training Accuracy: {final_train_acc * 100:.2f}%")  
print(f"Final Validation Accuracy: {final_val_acc * 100:.2f}%")
```

... Final Training Accuracy: 95.83%
Final Validation Accuracy: 99.00%

Technologies to be used

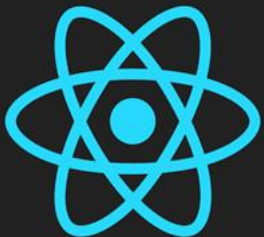


Keras

A deep learning library



TensorFlow



React Native



Firebase



GitHub

Functional Requirements

- Users can upload or capture real-time facial images.
- Track emotional trends over time.
- Identify and classify emotions

Non-Functional Requirements

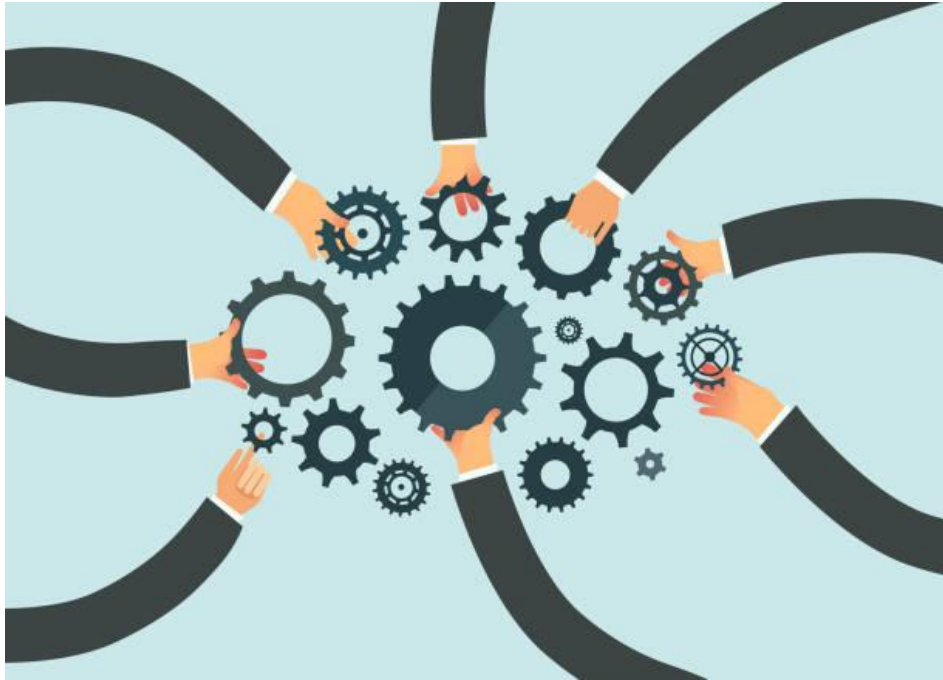
- Handle many users and allow future updates.
- Detect emotions within 2 seconds with 90%+ accuracy.



Supporting Items (Commercialization)

Target Market:

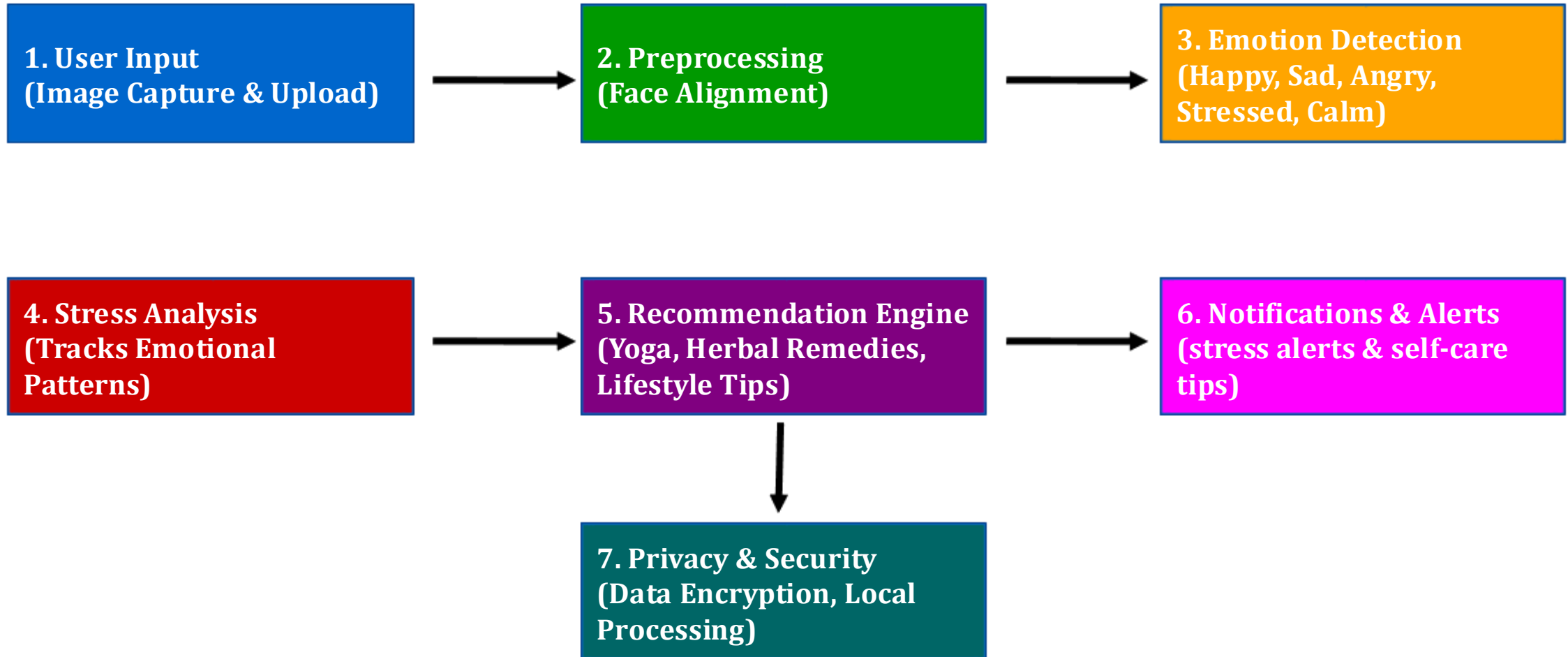
- Health-conscious individuals.
- Urban and rural Ayurveda enthusiasts.



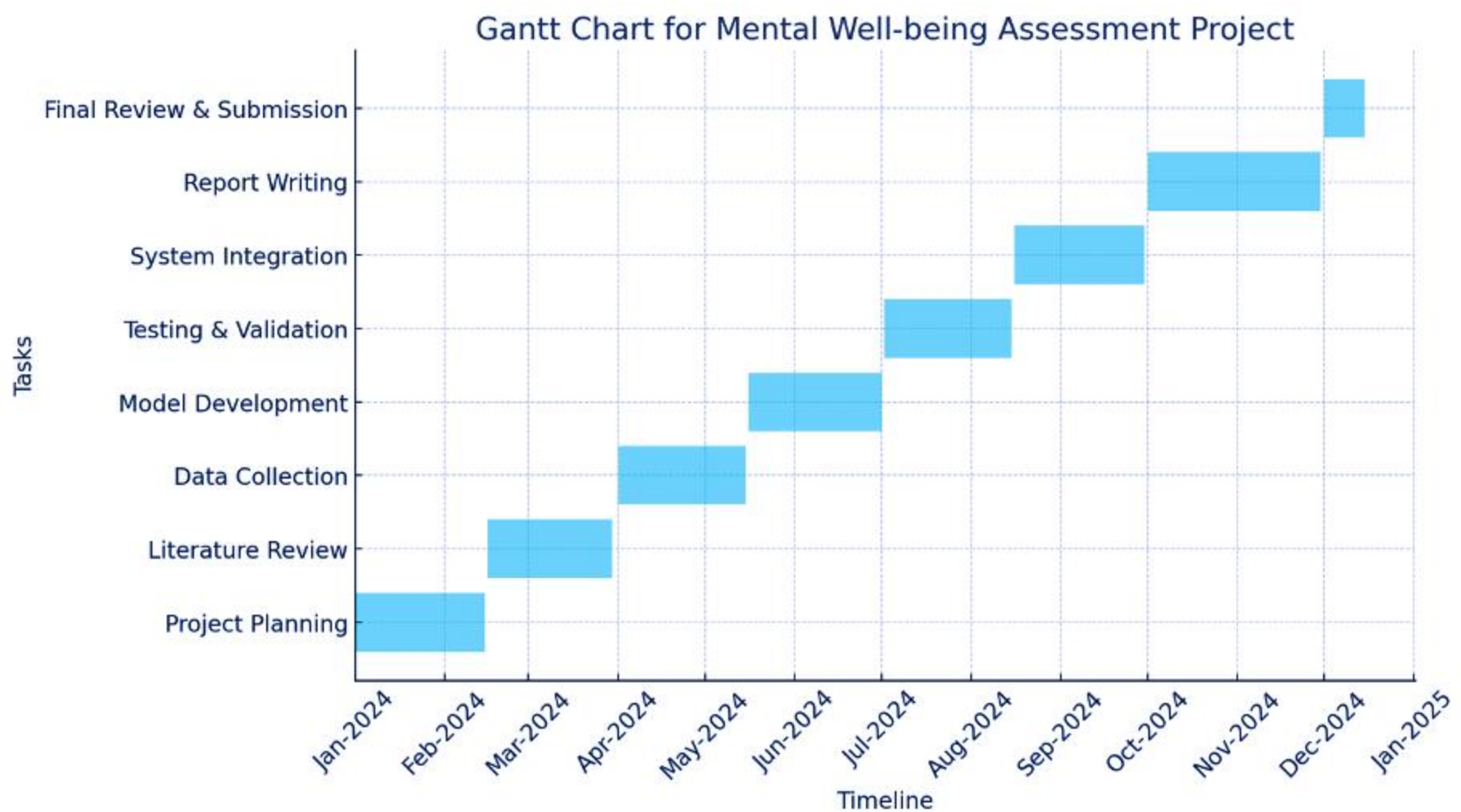
Partnerships:

- Collaborate with Ayurvedic practitioners and wellness centers.
- Partner with wearable tech companies for seamless integration.

System Diagram



Gantt Chart



Work Breakdown Chart

Project Planning & Requirements Gathering

- Define project scope, objectives, and team roles
- Develop project timeline and milestones

Data Collection & Preprocessing

- Identify data sources
- Collect facial expression dataset
- Preprocess and clean dataset

AI Model Development

- Select suitable model architecture
- Train emotion detection model
- Evaluate model performance

System Development

- Develop web/mobile application
- Integrate AI model with system
- Ensure real-time processing

Testing & Validation

- Test system with sample users
- Gather and analyze feedback
- Optimize system performance

Deployment & Maintenance

- Deploy final system
- Develop user manuals and tutorials
- Monitor system performance

REFERENCES

- Lad, Vasant. Textbook of Ayurveda: Fundamental Principles of Ayurveda. Ayurvedic Press, 2002.
- Yang, Guang-Zhong, et al. "The future of wearable technologies for personalized healthcare." Nature Electronics 1.6 (2018): 302-312.
- McCall, Timothy. Yoga as Medicine: The Yogic Prescription for Health and Healing. Bantam, 2007.
- Bonato, Paolo. "Wearable sensors and systems." IEEE Engineering in Medicine and Biology Magazine 29.3 (2010): 25-36.

CONCLUSION

- This system combines AI-driven emotion detection with Ayurvedic well-being to provide real-time stress monitoring and personalized self-care.
- ensures accuracy, privacy, and accessibility, empowering users to manage their mental health effectively.





IT21190520 | KAUSHALYA H.G.B

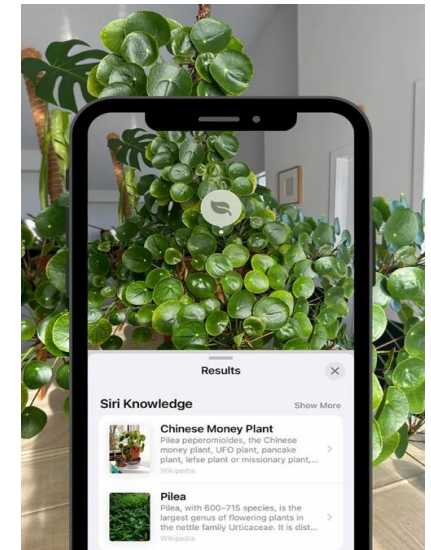
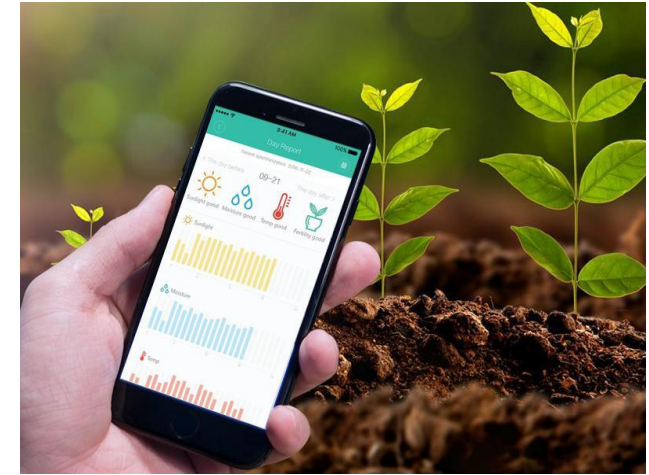
**AI-BASED AYURVEDIC MEDICINAL PLANT IDENTIFICATION
AND DISEASE DIAGNOSIS WITH TREATMENT
RECOMMENDATION USING LEAF IMAGE ANALYSIS.**

Information Technology (IT)

Introduction

BACKGROUND

- **Accurately identify Ayurvedic medicinal plants** through real-time image analysis.
- **Diagnose the health condition** of the leaf—distinguishing between healthy and diseased states.
- **Receive personalized Ayurvedic treatment suggestions** and disease control guidance for affected plants.



Modified Function: Plant Identification

- Processes static images of medicinal plant leaves for high accuracy.
- Identifies plants used for five specific diseases.
- Provides essential details: plant characteristics, medicinal benefits, and usage guidance.
- Detects leaf health by distinguishing between healthy and diseased leaves.
- Offers tailored Ayurvedic treatment recommendations and disease control advice.

Selected Plants, Their Traditional Uses & Leaf Health Guidelines

- **බෙලි (Beli) — *Aegle marmelos***
 - **Traditional Use:**
 - Used to treat gastrointestinal disorders such as diarrhea and dysentery.
 - Supports digestive health through antioxidant-rich decoctions.
 - **Leaf Health Guidelines:**
 - Healthy leaves provide reliable therapeutic benefits.
 - Diseased leaves (e.g., discoloration, spotting) may indicate fungal or bacterial issues and should be discarded to avoid reduced efficacy.
- **නික (Nika) — *Vitex negundo***
 - **Traditional Use:**
 - Employed as an anti-inflammatory agent to reduce fever, joint pain, and arthritis symptoms.
 - Often used in poultices or decoctions for its analgesic properties.
 - **Leaf Health Guidelines:**
 - Only fresh, healthy leaves ensure effective treatment.
 - Leaves showing signs of disease (e.g., lesions or mold) should be avoided as they may compromise the remedy.
- **ආවට්ටා — *Justicia adhatoda***
 - **Traditional Use:**
 - Commonly used for respiratory conditions like cough, asthma, and bronchitis due to its expectorant properties.
 - The leaves, when used in decoctions, help alleviate congestion.
 - **Leaf Health Guidelines:**
 - Only healthy, uncontaminated leaves should be used to guarantee safety and potency.
 - Diseased leaves may harbor pathogens that can impair the treatment's effectiveness and should be replaced.

Research Gap

Features Comparison Table

Features	Research 1	Research 2	Research 3	Research 4	Proposed Solution
User-Friendly Interface	Low	Low	Medium	Low	High
Mobile Application	✗	✗	✓	✗	✓
AI-Based Medicinal Plant Identification	✗	✗	✗	✗	✓
Leaf Image Processing for Identification	✗	✗	✗	✗	✓
Database of Ayurvedic Medicinal Plants	✗	✗	✓	✗	✓
Disease-Based Plant Recommendation	✗	✗	✗	✗	✓
Offline Accessibility	✗	✗	✗	✗	✓
Integration with Ayurvedic Treatment Guidance	✗	✗	✓	✗	✓
Machine Learning-Based Accuracy Improvement	✗	✗	✓	✗	✓

Research Problems

- Existing AI models show lower accuracy in identifying Ayurvedic medicinal plants.
- A lack of structured, high-quality datasets hampers model performance.
- Variations in leaf morphology and seasonal changes increase difficulty in accurate detection.
- Users often remain unaware of the precise medicinal uses of the identified plants.
- Real-time plant identification is challenging in rural areas with limited connectivity.
- Current systems do not connect the identified plant profiles to user-specific treatment recommendations.



Research Objective



Primary Objective:

- Develop an AI-powered plant identification system that accurately recognizes medicinal plant leaves and delivers personalized Ayurvedic treatment guidance for selected diseases.

Specific Goals:

- Develop and train an accurate AI model for plant identification.
- Create a comprehensive medicinal plant database with Ayurvedic treatments.
- Enable offline accessibility for rural users.
- Integrate personalized Ayurvedic guidance into the app.
- Design a user-friendly mobile interface for easy plant recognition.
- Enhance model accuracy with diverse leaf images.

Achievement



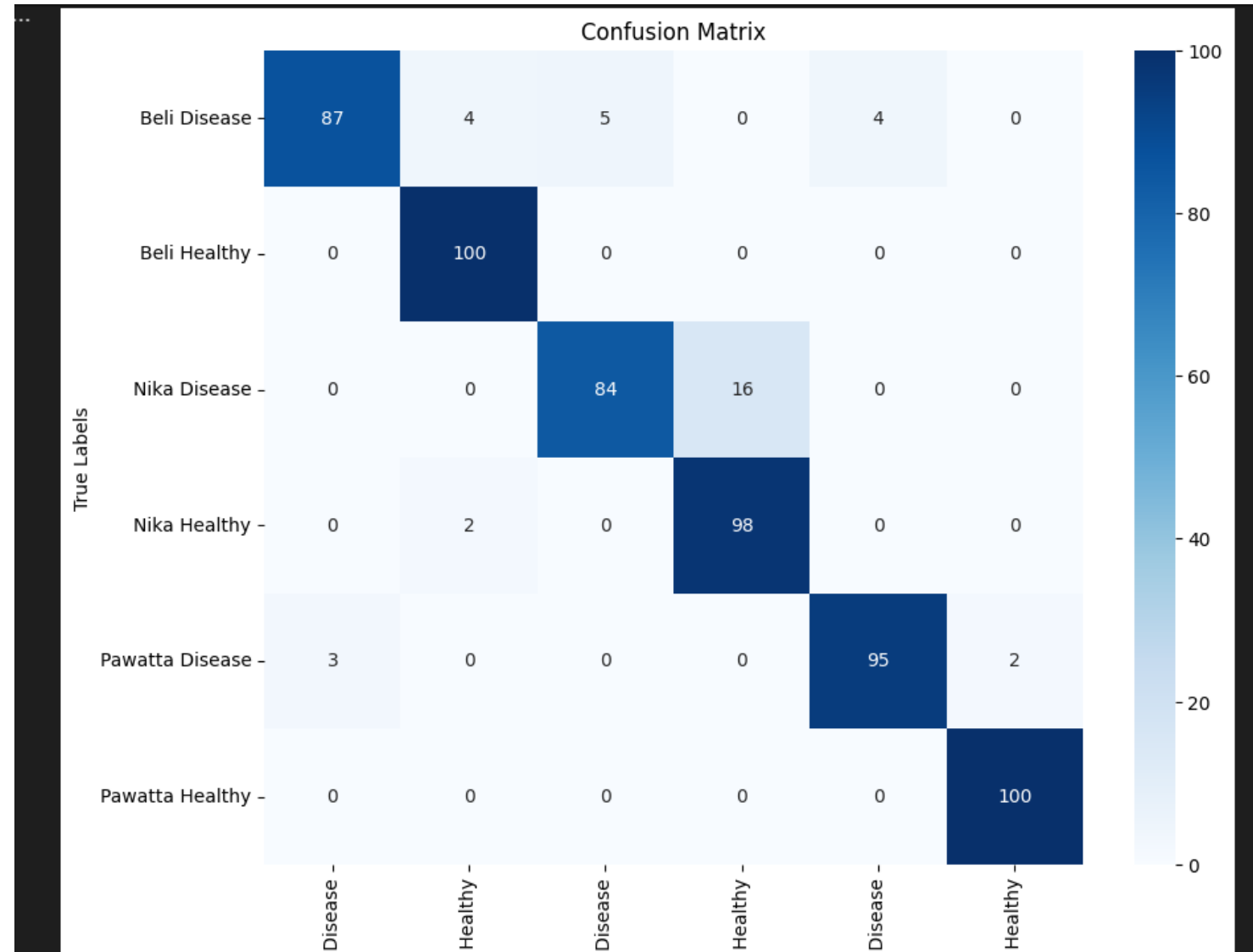
- Successfully trained an AI model that accurately identifies Ayurvedic medicinal plants using leaf images.
- Established a comprehensive database linking identified plants to their Ayurvedic treatment practices for five selected diseases.
- Implemented a novel module that distinguishes between healthy and diseased leaves, triggering tailored treatment recommendations.

Designed and piloted a user-friendly mobile interface with offline functionality to ensure accessibility in rural areas.

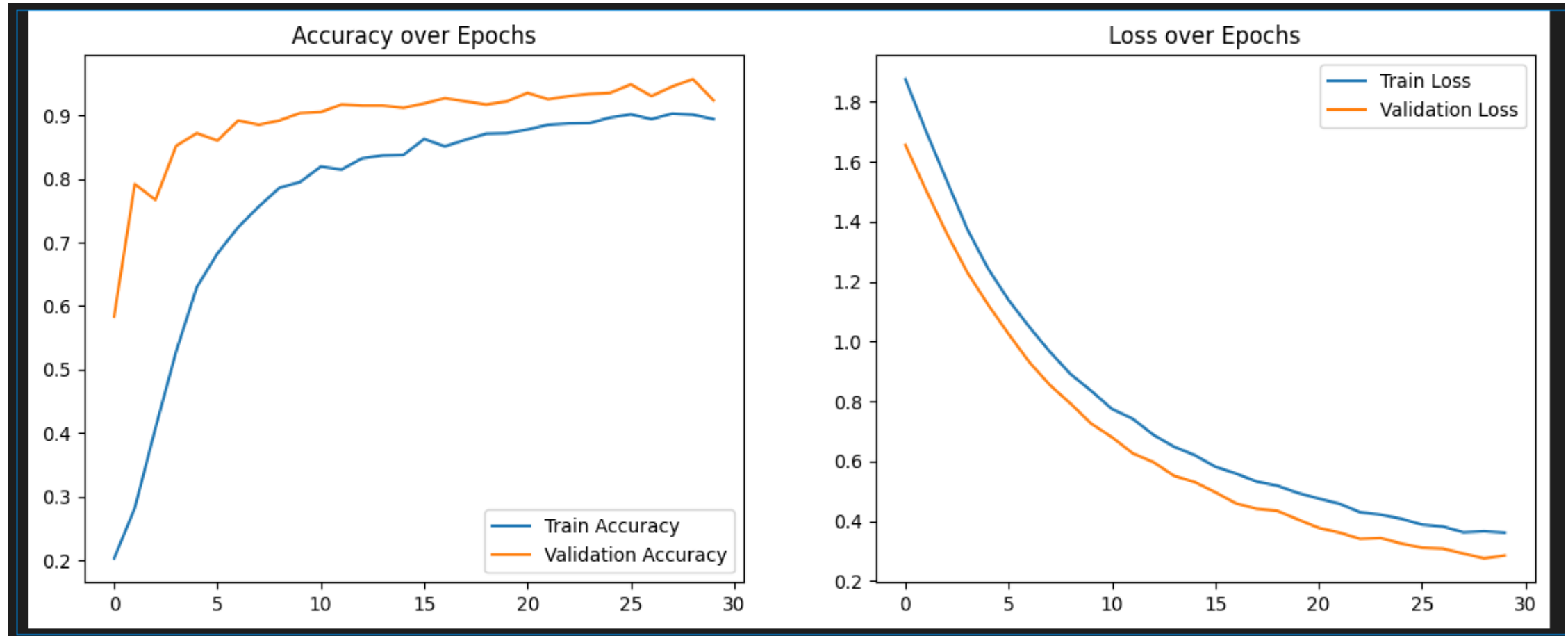
Confusion Matrix

... Classification Report:

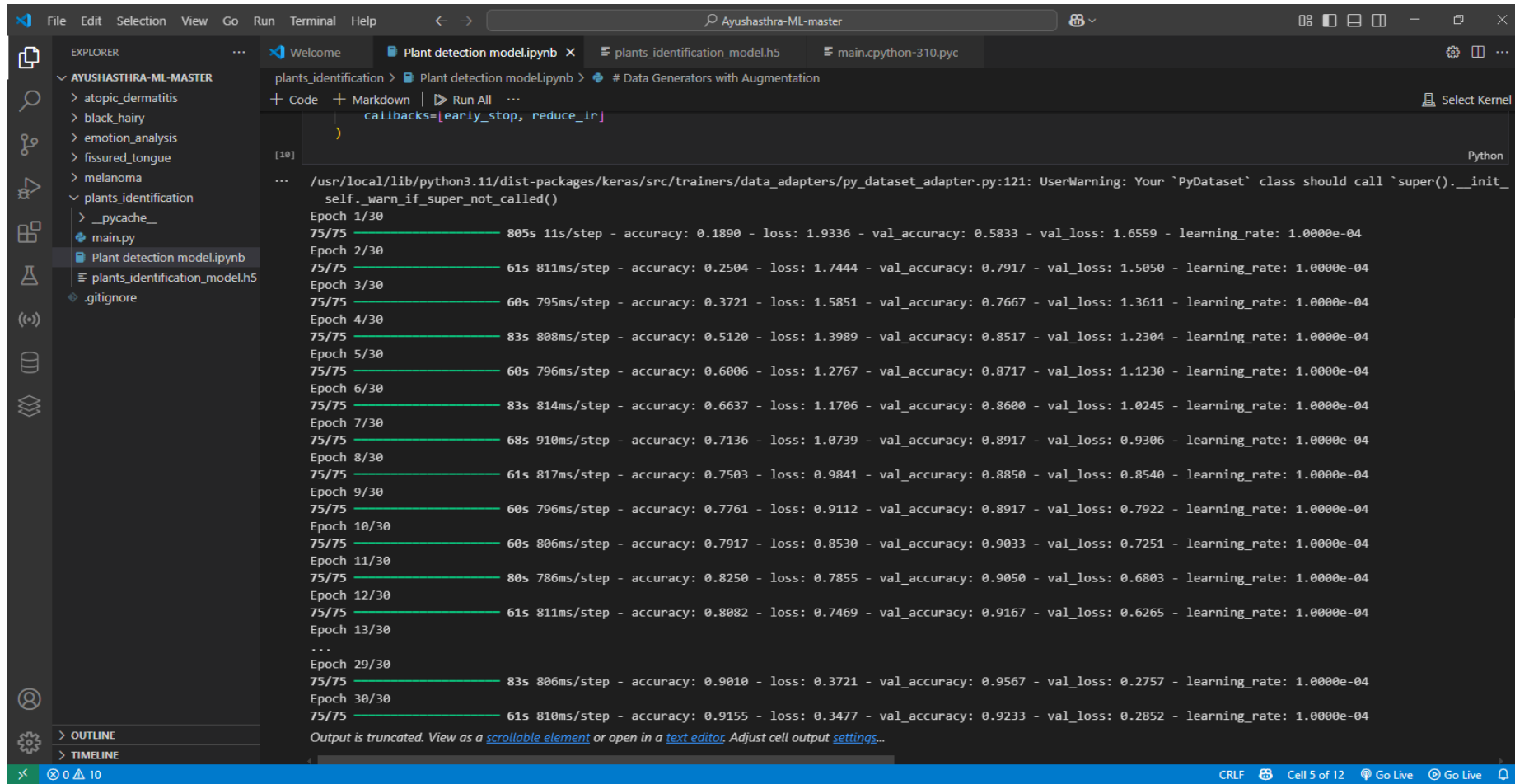
	precision	recall	f1-score	support
Beli Disease	0.97	0.87	0.92	100
Beli Healthy	0.94	1.00	0.97	100
Nika Disease	0.94	0.84	0.89	100
Nika Healthy	0.86	0.98	0.92	100
Pawatta Disease	0.96	0.95	0.95	100
Pawatta Healthy	0.98	1.00	0.99	100
accuracy			0.94	600
macro avg	0.94	0.94	0.94	600
weighted avg	0.94	0.94	0.94	600



Testing Validation Accuracy & Lost



Accuracy – 92%



```
plants_identification > Plant detection model.ipynb > # Data Generators with Augmentation
+ Code + Markdown | ▶ Run All ...
callbacks=[early_stop, reduce_lr]

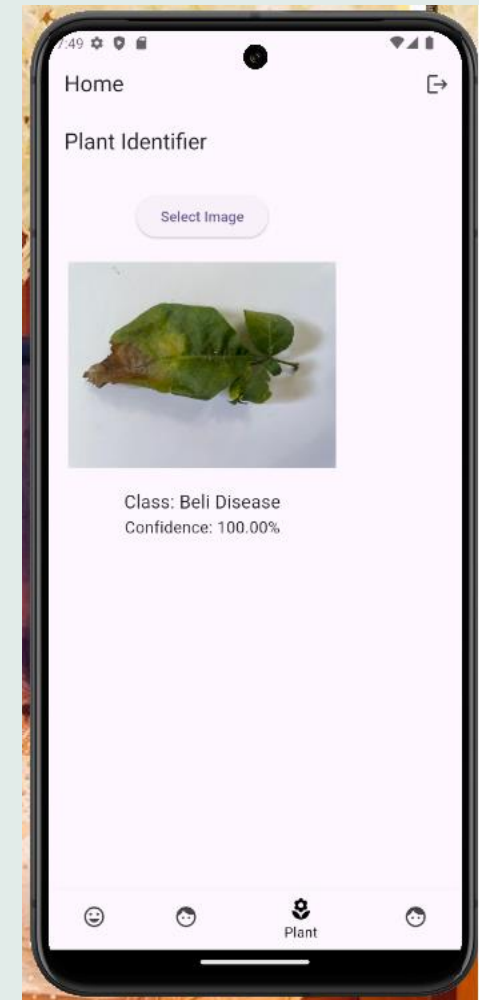
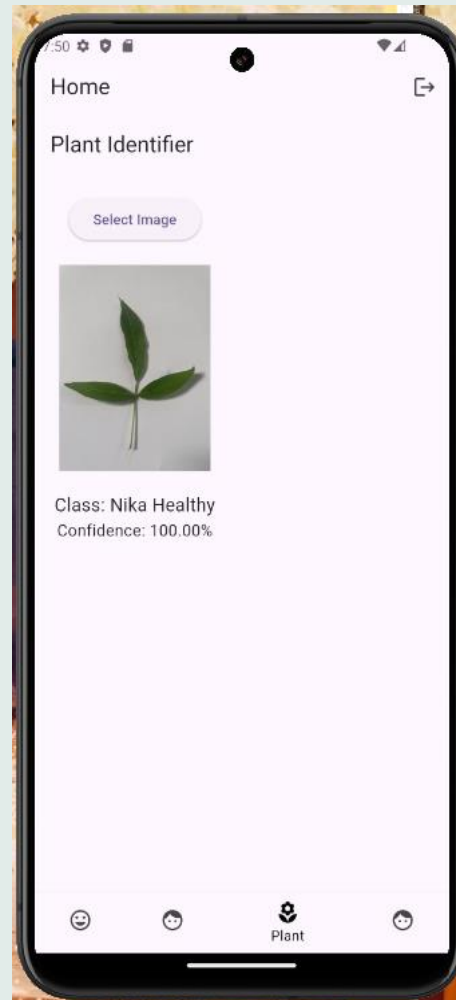
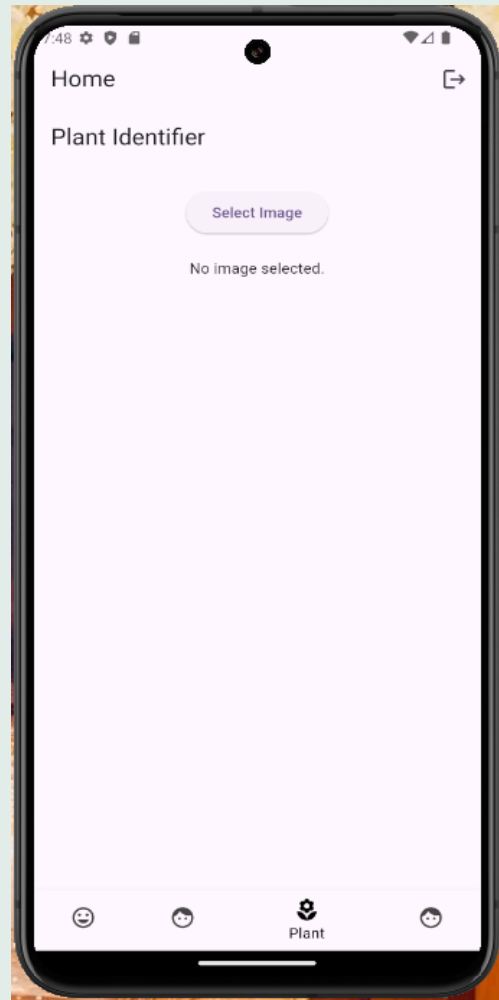
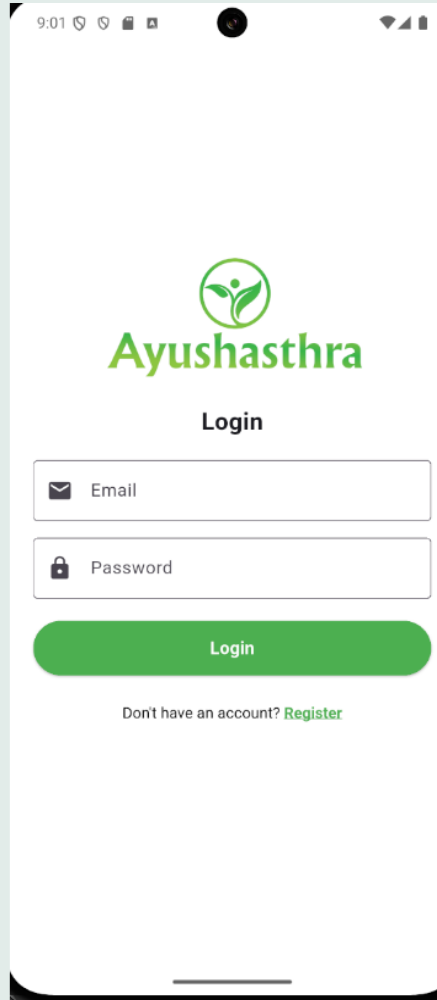
[10] Python

... /usr/local/lib/python3.11/dist-packages/keras/src/trainers/data_adapters/py_dataset_adapter.py:121: UserWarning: Your `PyDataset` class should call `super().__init__`
self._warn_if_super_not_called()
Epoch 1/30
75/75 — 805s 11s/step - accuracy: 0.1890 - loss: 1.9336 - val_accuracy: 0.5833 - val_loss: 1.6559 - learning_rate: 1.0000e-04
Epoch 2/30
75/75 — 61s 81ms/step - accuracy: 0.2504 - loss: 1.7444 - val_accuracy: 0.7917 - val_loss: 1.5050 - learning_rate: 1.0000e-04
Epoch 3/30
75/75 — 60s 79ms/step - accuracy: 0.3721 - loss: 1.5851 - val_accuracy: 0.7667 - val_loss: 1.3611 - learning_rate: 1.0000e-04
Epoch 4/30
75/75 — 83s 88ms/step - accuracy: 0.5120 - loss: 1.3989 - val_accuracy: 0.8517 - val_loss: 1.2304 - learning_rate: 1.0000e-04
Epoch 5/30
75/75 — 60s 79ms/step - accuracy: 0.6006 - loss: 1.2767 - val_accuracy: 0.8717 - val_loss: 1.1230 - learning_rate: 1.0000e-04
Epoch 6/30
75/75 — 83s 84ms/step - accuracy: 0.6637 - loss: 1.1706 - val_accuracy: 0.8600 - val_loss: 1.0245 - learning_rate: 1.0000e-04
Epoch 7/30
75/75 — 68s 91ms/step - accuracy: 0.7136 - loss: 1.0739 - val_accuracy: 0.8917 - val_loss: 0.9306 - learning_rate: 1.0000e-04
Epoch 8/30
75/75 — 61s 81ms/step - accuracy: 0.7503 - loss: 0.9841 - val_accuracy: 0.8850 - val_loss: 0.8540 - learning_rate: 1.0000e-04
Epoch 9/30
75/75 — 60s 79ms/step - accuracy: 0.7761 - loss: 0.9112 - val_accuracy: 0.8917 - val_loss: 0.7922 - learning_rate: 1.0000e-04
Epoch 10/30
75/75 — 60s 80ms/step - accuracy: 0.7917 - loss: 0.8530 - val_accuracy: 0.9033 - val_loss: 0.7251 - learning_rate: 1.0000e-04
Epoch 11/30
75/75 — 80s 78ms/step - accuracy: 0.8250 - loss: 0.7855 - val_accuracy: 0.9050 - val_loss: 0.6803 - learning_rate: 1.0000e-04
Epoch 12/30
75/75 — 61s 81ms/step - accuracy: 0.8082 - loss: 0.7469 - val_accuracy: 0.9167 - val_loss: 0.6265 - learning_rate: 1.0000e-04
Epoch 13/30
...
Epoch 29/30
75/75 — 83s 80ms/step - accuracy: 0.9010 - loss: 0.3721 - val_accuracy: 0.9567 - val_loss: 0.2757 - learning_rate: 1.0000e-04
Epoch 30/30
75/75 — 61s 81ms/step - accuracy: 0.9155 - loss: 0.3477 - val_accuracy: 0.9233 - val_loss: 0.2852 - learning_rate: 1.0000e-04
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

Successful Data Collection for Model Training



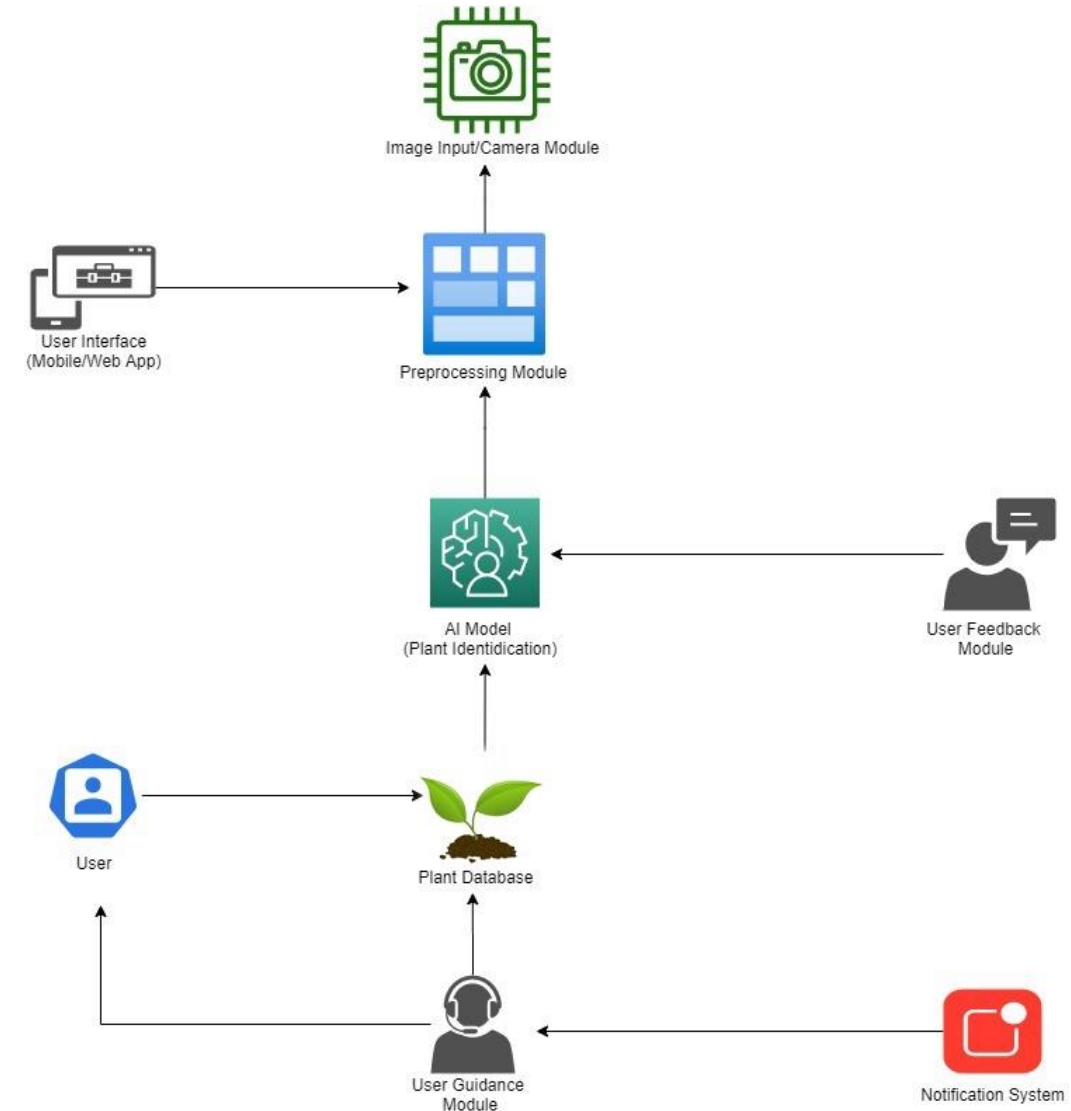
Ui design prototype



METHODOLOGY

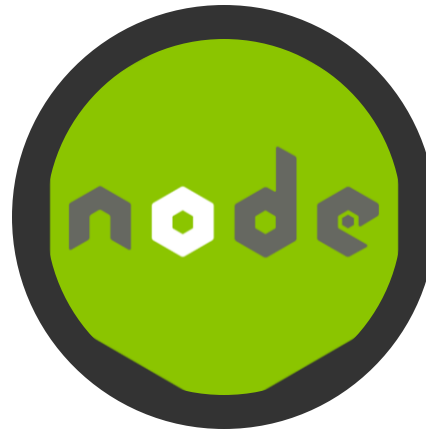
System Diagram

- **1. Leaf Image Capture:**
The user uploads a clear image of the medicinal plant leaf via the mobile app.
- **2. Image Preprocessing:**
The system enhances the image quality and segments the leaf to focus on critical features.
- **3. AI-Based Plant Identification:**
The refined image is analyzed by the AI model, which accurately determines the plant species.
- **4. Result Display:**
The app presents the plant name, detailed medicinal properties, usage guidance, and—in cases where the leaf appears diseased—tailored treatment recommendations and preventive measures.

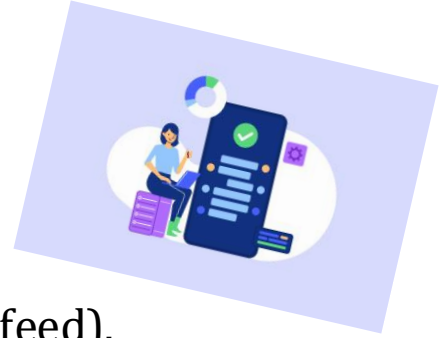


Technologies to be used

- Machine Learning – PyTouch
- Backend – node.js
- Mobile app development- flutter
- Image Processing – OpenCV
- Model Training – Google collabs
- Database- MongoDB



Requirements Specification



Functional Requirements

- The system shall identify medicinal plants based on uploaded static leaf images (not live camera feed).
- The system shall maintain a database of Ayurvedic medicinal plants linked to five selected diseases.
- The app shall display plant names, medicinal uses, and Ayurvedic treatment guidance.
- The system shall allow offline access to identification and treatment data.

Non-Functional Requirements

- **Scalability:** The system must support increasing numbers of users and plant species without performance degradation.
- **Performance:** The AI model must process leaf images and return results within a few seconds.
- **Security:** All user data and plant information must be securely stored with proper privacy controls.
- **Usability:** The app interface must be user-friendly and intuitive for both tech-savvy and non-technical users.
- **Offline Accessibility:** Core functionalities (identification + treatment guidance) must work without internet access.
- **Maintainability:** The system should be modular and easy to update with new plant data and treatments.
- **Compatibility:** The app must be compatible with both Android and iOS mobile platforms

WBS - AI-Powered Plant Identification and Usage Guidance System

Data Collection

Collect plant images in real-time

Annotate images with plant names and medicinal properties

Ensure data quality and consistency

Store data securely

AI Model Development

Design and build an AI model for plant identification

Train and test the model with the dataset

Optimize for speed and accuracy

Plant Database Creation

Develop a comprehensive database of Ayurvedic plants

Link each plant to its medicinal benefits and applications

Integrate real-time plant identification with the database

User Interface Design

Design a user-friendly interface for plant image uploads

Display real-time identification results

Provide easy-to-understand usage guidance based on the plant

Usage Guidance Module

Provide users with Ayurvedic treatments based on plant identification

Suggest applications and usage based on medical conditions

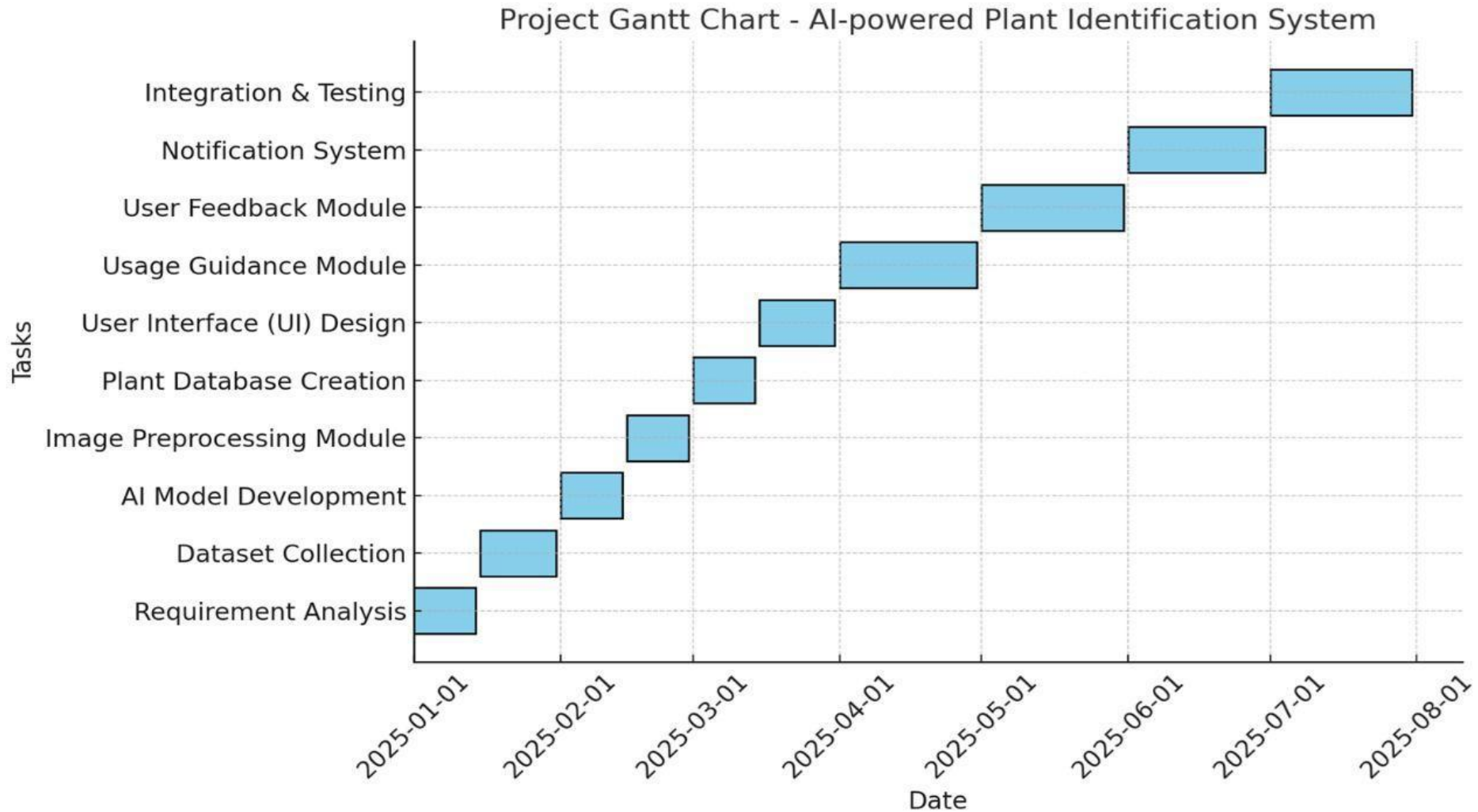
Testing and Validation

Test for accuracy with real plant images

Validate guidance with Ayurvedic experts

Collect feedback and improve the system

Gantt Chart



Supporting Items (Commercialization)

Target Market:

- To assist in accurate plant identification and quick reference to traditional treatments.
- Especially those in rural areas, who rely on medicinal plants for natural remedies and require offline access.
- Studying botany, Ayurveda, or plant-based medicine, looking for a smart reference tool.
- Interested in using natural treatments and identifying medicinal plants around them.
- For identifying and sourcing correct plant species for production of herbal remedies and oils.



Partnerships:

- To validate plant identification and treatment accuracy based on traditional knowledge.
- For access to verified plant datasets and expert insights on medicinal plant cultivation.
- Collaborate with AI developers and mobile app experts to enhance model performance and app functionality.
- Especially those working in rural healthcare and herbal education, to help distribute the app to underserved communities.
- For real-world feedback and potential integration with herbal product recommendations.

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CONCLUSION

- Combines AI with Ayurvedic knowledge for smart plant identification.
- Detects plant species and leaf health using image processing.
- Provides personalized Ayurvedic treatment guidance.
- Supports offline use, making it ideal for rural communities.
- Promotes accessible, sustainable herbal health practices.



